

TECHNICAL SPECIFICATION



UHV AC transmission systems – Part 201: UHV AC substation design



THIS PUBLICATION IS COPYRIGHT PROTECTED

Copyright © 2018 IEC, Geneva, Switzerland

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester. If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
info@iec.ch
www.iec.ch

About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

IEC Catalogue - webstore.iec.ch/catalogue

The stand-alone application for consulting the entire bibliographical information on IEC International Standards, Technical Specifications, Technical Reports and other documents. Available for PC, Mac OS, Android Tablets and iPad.

IEC publications search - webstore.iec.ch/advsearchform

The advanced search enables to find IEC publications by a variety of criteria (reference number, text, technical committee,...). It also gives information on projects, replaced and withdrawn publications.

IEC Just Published - webstore.iec.ch/justpublished

Stay up to date on all new IEC publications. Just Published details all new publications released. Available online and also once a month by email.

Electropedia - www.electropedia.org

The world's leading online dictionary of electronic and electrical terms containing 21 000 terms and definitions in English and French, with equivalent terms in 16 additional languages. Also known as the International Electrotechnical Vocabulary (IEV) online.

IEC Glossary - std.iec.ch/glossary

67 000 electrotechnical terminology entries in English and French extracted from the Terms and Definitions clause of IEC publications issued since 2002. Some entries have been collected from earlier publications of IEC TC 37, 77, 86 and CISPR.

IEC Customer Service Centre - webstore.iec.ch/csc

If you wish to give us your feedback on this publication or need further assistance, please contact the Customer Service Centre: sales@iec.ch.



IEC TS 63042-201

Edition 1.0 2018-12

TECHNICAL SPECIFICATION



**UHV AC transmission systems –
Part 201: UHV AC substation design**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

ICS 29.240.01; 29.240.10

ISBN 978-2-8322-6330-3

Warning! Make sure that you obtained this publication from an authorized distributor.

CONTENTS

FOREWORD.....	6
1 Scope	8
2 Normative references	8
3 Terms and definitions	9
4 UHV AC substation requirement	10
4.1 General requirement	10
4.2 System demands	10
4.3 Operation and maintenance requirements	11
4.4 Construction requirements	11
4.5 Site condition	11
4.6 Environmental impact.....	12
4.7 Economy.....	12
5 Bus scheme and feeder connection	12
5.1 General.....	12
5.2 Scheme at high-voltage side of main transformer.....	13
5.3 Scheme at intermedium-voltage side of main transformer	14
5.4 Scheme at low-voltage side of main transformer	14
5.5 System neutral earthing mode of a UHV AC substation	15
6 Selection of equipment and conductors	16
6.1 General.....	16
6.1.1 Voltage	16
6.1.2 Rated current.....	16
6.1.3 Rated frequency	16
6.2 Basic requirements	16
6.2.1 Electrical requirements	16
6.2.2 Mechanical requirements	17
6.2.3 Environmental conditions.....	17
6.3 Transformer	18
6.4 UHV shunt reactor and neutral-earthing reactor	19
6.5 UHV switchgear	19
6.6 UHV circuit breaker.....	20
6.7 UHV disconnecter	20
6.8 UHV earthing switch for maintenance	21
6.9 High-speed earthing switch	22
6.10 UHV current transformer	22
6.11 UHV voltage transformer.....	22
6.12 UHV surge arrester	23
6.13 Reactive power compensation device for low voltage side of UHV transformer	23
6.14 UHV bushing.....	23
6.15 UHV insulator	24
6.16 UHV conductor.....	24
6.16.1 General	24
6.16.2 Conductor type	24
6.16.3 Selection of current-carrying capacity (cross-section)	25
6.16.4 Corona and radio interference	25
6.16.5 Mechanical strength.....	26

7	Equipment layout.....	26
7.1	General requirement of equipment layout.....	26
7.1.1	General	26
7.1.2	Optimization of substation layout	26
7.1.3	Seismic performance	26
7.1.4	Construction, serviceability and reliability and failure response ability.....	26
7.2	Minimum clearances	27
7.2.1	Normal environmental conditions	27
7.2.2	Minimum clearances in air-voltage range	27
7.3	Electromagnetic environment	27
7.3.1	Electrostatic induction mitigation design	27
7.3.2	Magnetic induction mitigation design	28
7.3.3	Audible noise mitigation design	28
7.4	Selection of switchgear equipment.....	29
7.5	Switchgear Installations layout.....	29
7.5.1	General	29
7.5.2	Location arrangement of switchgear	29
7.5.3	Basic arrangement of surge arresters	30
7.5.4	Optimal gas-insulated busbar (GIB) layout and length	30
7.5.5	Utilization of working space for substations	30
7.6	Protection against direct lightning strike.....	31
7.7	Earthing systems	31
7.7.1	General considerations	31
7.7.2	Multiple point earthing method for GIS	32
7.8	Seismic design.....	33
7.8.1	General	33
7.8.2	Basic seismic design	33
7.8.3	Special seismic performance for UHV AC substation equipment	34
8	Control, protection and communication	35
8.1	General.....	35
8.2	Control system.....	35
8.3	Relay protection.....	36
8.3.1	General	36
8.3.2	Duplicated configuration of UHV AC equipment relay protection	36
8.3.3	UHV transformer protection	36
8.4	Communication	37
8.5	Electromagnetic compatibility requirements for control and protection equipment.....	37
9	DC and AC auxiliary power supply system	38
9.1	General.....	38
9.2	DC power supply system.....	38
9.3	AC auxiliary power supply system	38
9.4	AC uninterruptible power supply (UPS) system	39
10	UHV gantry, support and foundation design	39
10.1	UHV gantry and support design	39
10.1.1	General	39
10.1.2	Load and combination of loads	39
10.1.3	Detailing requirements.....	40
10.2	GIS or MTS foundation design	41

Annex A (informative) Load combination of UHV AC equipment	43
Annex B (informative) Specification of UHV AC equipment and conductor	44
Annex C (informative) 1 000 kV outdoor overhead flexible conductor for UHV AC substations in China	46
C.1 General.....	46
C.2 Environmental conditions	46
C.3 Current-carrying capacity and thermal stability check.....	46
C.3.1 Current-carrying capacity check.....	46
C.3.2 Thermal stability check	47
C.4 Determination of bundle spacing	48
C.4.1 General	48
C.4.2 Calculation of maximum electric field strength around conductor	48
C.5 Corona inception voltage	49
C.6 Electric field strength on ground caused by electrostatic induction	50
Annex D (informative) Corona noise reduction measures of a UHV AC substation conductor under the rainy condition in Japan	52
D.1 Basic concept of corona noise reduction	52
D.2 Structure design of UHV AC substation conductor	52
D.3 Design criteria of partial discharge on UHV AC substation conductor	54
D.4 Corona noise measurement of the entrance in UHV AC test station	54
Annex E (informative) Typical examples of items to be considered to select switchgear type	56
Annex F (informative) Standards related to seismic design	58
F.1 Typical seismic guide and standards.....	58
F.2 Comparison of main items among the seismic standards	58
Bibliography.....	59
 Figure 1 – Birds eye view of a typical UHV AC substation	10
Figure 2 – Double busbar (DB) with or without bus section connection.....	13
Figure 3 – One-and-a-half circuit breaker (OHCB)	14
Figure 4 – Two-circuit breaker (2CB)	14
Figure 5 – Example diagram of a bus scheme and feeder connection	15
Figure 6 – Typical configuration of UHV gas-insulated switchgear and crane location	31
Figure 7 – Earthing methods	33
Figure 8 – Flow chart for seismic qualification.....	34
Figure 9 – Example of continuous UHV gantry and independent gantry.....	41
Figure 10 – GIS foundation forms	42
Figure C.1 – Relationship between maximum electric field strength and bundle spacing	49
Figure C.1 – Layout plan of main transformer incoming lines	51
Figure D.1 – Conductor design of UHV AC substation.....	53
 Table 1 – Comparison of a four-legged reactor and HSES	22
Table 2 – Comparison of conductors.....	25
Table A.1 – Example of load combination for UHV AC equipment	43
Table B.1 – UHV voltage specification	44
Table B.2 – Specification of UHV short-circuit current.....	44

Table B.3 – Noise specification	44
Table B.4 – Surge arrester specification applied in different countries	45
Table C.1 – Current-carrying capacity of bundle conductor	47
Table C.2 – Corona inception voltage of conductor	50
Table D.1 – Estimated values of corona noise of UHV AC transmission line.....	53
Table D.2 – Design criteria of partial discharge on UHV AC substation conductor	54
Table D.3 – Results of corona noise measurements and average value of corona noise	55
Table E.1 – The principal technology designs for substations (CIGRE TB 570)	56
Table E.2 – Typical examples of items to be considered to select switchgear type	57
Table F.1 – Typical seismic guide and standards	58
Table F.2 – Comparison of main items among seismic standards.....	58

INTERNATIONAL ELECTROTECHNICAL COMMISSION

UHV AC TRANSMISSION SYSTEMS –**Part 201: UHV AC substation design****FOREWORD**

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

The main task of IEC technical committees is to prepare International Standards. In exceptional circumstances, a technical committee may propose the publication of a technical specification when

- the required support cannot be obtained for the publication of an International Standard, despite repeated efforts, or
- the subject is still under technical development or where, for any other reason, there is the future but no immediate possibility of an agreement on an International Standard.

Technical specifications are subject to review within three years of publication to decide whether they can be transformed into International Standards.

IEC TS 63042-201, which is a technical specification, has been prepared by IEC technical committee 122: UHV AC transmission systems.

The text of this Technical Specification is based on the following documents:

Enquiry draft	Report on voting
122/64/DTS	122/71A/RVDTS

Full information on the voting for the approval of this technical specification can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 63042 series, published under the general title *UHV AC transmission systems*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

UHV AC TRANSMISSION SYSTEMS –

Part 201: UHV AC substation design

1 Scope

This part of 63042, which is a Technical Specification, provides common rules for the design of substations with the highest voltages of AC transmission systems exceeding 800 kV, so as to provide safety and proper functioning for the intended use.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60038:2009, *IEC standard voltages*

IEC 60044 (all parts), *Instrument transformers*

IEC 60059:1999, *IEC standard current ratings*
IEC 60059:1999/AMD1:2009

IEC 60071-1:2006, *Insulation co-ordination – Part 1: Definitions, principles and rules*
IEC 60071-1:2006/AMD1:2010

IEC 60071-2, *Insulation co-ordination – Part 2: Application guide*

IEC 60076 (all parts), *Power transformers*

IEC 60068-3-3, *Environmental testing – Part 3: Guidance – Seismic test methods for equipments*

IEC 60137, *Insulated bushings for alternating voltages above 1000 V*

IEC 60168, *Tests on indoor and outdoor post insulators of ceramic material or glass for systems with nominal voltages greater than 1000 V*

IEC 60196:2009, *IEC standard frequencies*

IEC 60255-26, *Measuring relays and protection equipment – Part 26: Electromagnetic compatibility requirements*

IEC TS 60479-1, *Effects of current on human beings and livestock – Part 1: General aspects*

IEC 60721-2-4, *Classification of environmental conditions – Part 2-4: Environmental conditions appearing in nature – Solar radiation and temperature*

IEC TS 60815 (all parts), *Selection and dimensioning of high-voltage insulators intended for use in polluted conditions*

IEC 60865 (all parts), *Short-circuit currents*

IEC 60871 (all parts), *Shunt capacitors for a.c. power systems having a rated voltage above 1 000 V*

IEC 60909 (all parts), *Short-circuit currents in three-phase a.c. systems*

IEC TS 61463, *Bushings – Seismic qualification*

IEC 61850 (all parts), *Communication networks and systems for power utility automation*

IEC 61936-1:2010, *Power installations exceeding 1 kV a.c. – Part 1: Common rules*
IEC 61936-1:2010/AMD1:2014

IEC 62231, *Composite station post insulators for substations with AC voltages greater than 1 000 V up to 245 kV – Definitions, test methods and acceptance criteria*

IEC 62271-100, *High-voltage switchgear and controlgear – Part 100: Alternating current circuit-breakers*

IEC 62271-102, *High-voltage switchgear and controlgear – Part 102: Alternating current disconnectors and earthing switches*

IEC 62271-207, *High-voltage switchgear and controlgear – Part 207: Seismic qualification for gas-insulated switchgear assemblies for rated voltages above 52 kV*

IEC TR 62271-300, *High-voltage switchgear and controlgear – Part 300: Seismic qualification of alternating current circuit-breakers*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1

UHV AC

highest voltage of AC transmission system exceeding 800 kV

Note 1 to entry: UHV stands for "ultra high voltage".

3.2

high-voltage side of transformer

highest voltage among two or three voltages on each side of the main transformer

3.3

intermedium voltage side of transformer

second highest voltage among three voltages on each side of the main transformer

3.4

low-voltage side of transformer

lowest voltage among two or three voltages in the apparatus or installation

Note 1 to entry: In this document, the definition is modified as the lowest voltage among two or more voltages on each side of main transformer.

4 UHV AC substation requirement

4.1 General requirement

The UHV AC substation should withstand the electrical, mechanical, climatic and environmental influences anticipated on site.

In the design of the UHV AC substation, several factors should be taken into account: system demands, operation and maintenance requirements, construction requirements, site condition, environmental impact and economy.

To meet the requirements above, the design should also be performed in consideration of the following characteristics of the UHV AC substation:

- high importance of the UHV AC substation;
- high transmission capacity;
- high overvoltage;
- high secondary arc current;
- high electric and magnetic field strength;
- large dimension and weight of the UHV AC equipment;
- large dimension of the UHV AC substation.

A typical design of a UHV AC substation is shown in Figure 1.

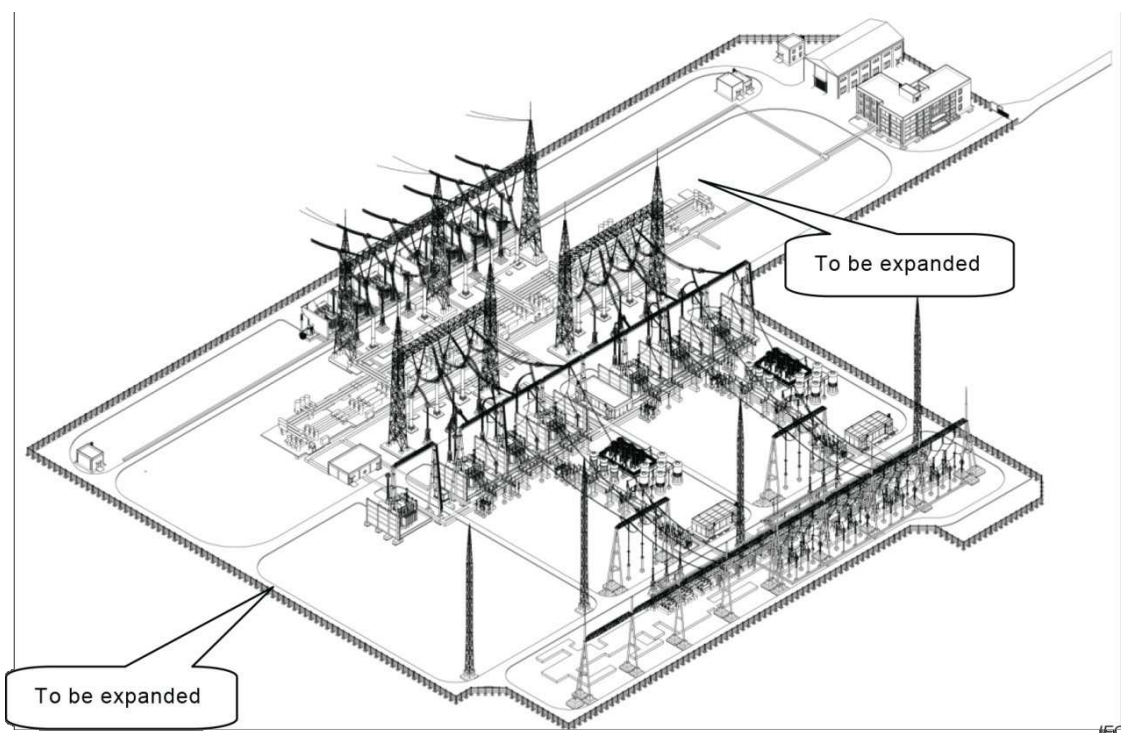


Figure 1 – Bird's-eye view of a typical UHV AC substation

4.2 System demands

The design of the UHV AC substation should meet the AC transmission system demands, including the following items:

- a) reliability: the bus scheme and feeder connection, equipment and control protection system of the UHV AC substation should be adopted to meet the requirements of reliability, considering the importance of the UHV AC substation in the AC transmission systems.
- b) availability: the selection of the bus scheme and feeder connection, the type and parameters of equipment, equipment layout, control protection and communication system should meet the demands of AC transmission system scale and parameters in the steady-state and transient-state and the possibility of extension (if required). Especially, the demands of the AC transmission system access should be taken into consideration in the design of equipment layout.
- c) flexibility: the convenience of switching feeders such as a bank of transformers or a transmission line, switching devices such as circuit breakers, busbars and their relay protection devices in case of maintenance, and the transition from the initial bus scheme and feeder connection to the final bus scheme and feeder connection, should be taken into consideration in the design of bus scheme and feeder connection as well as equipment layout.

4.3 Operation and maintenance requirements

The UHV AC substation with high applied voltage and large dimension plays a significant role in AC transmission systems, which makes the operation and maintenance of these substations very severe. The design of the UHV AC substation should meet the requirements of operation and maintenance, including the following items:

- a) The type of bus scheme and feeder connection and equipment should meet the requirements of operation and maintenance as much as possible on premise of meeting the system demands.
- b) The design of the UHV AC substation should limit the safety distance and electric field strength to a permissible value in order to ensure the operator safety and to meet the requirements of a daily operation and maintenance, even though the applied voltage of the UHV AC substation is high.
- c) The design of the earthing system should meet the requirements of a daily operation and maintenance. Especially, the step potential and touch potential of the earthing system should be limited to permissible value.
- d) Control, protection and communication system should meet the requirements of the system architecture and operation mode of AC transmission systems.

4.4 Construction requirements

Large and heavy equipment complicates the construction of the UHV AC substation. The design of the UHV AC substation should meet the requirements of construction, including the following items:

- a) The equipment layout and the distance between equipment should meet the requirements of transportation, lifting and installation of the equipment.
- b) When selecting the UHV conductors, the effort involved in the structural design shall be taken into account, too, besides the electrical properties such as current-carrying capacity, thermal stability, corona effect, etc.
- c) The transportation condition should meet the requirements for large equipment transport.
- d) The construction measures should be adopted to meet the requirements of the UHV AC equipment's foundation. Structure construction should be used in order to avoid the temperature cracks generated by the hydration heat impact in mass concrete construction.

4.5 Site condition

The design of the UHV AC substation should meet the requirements of the site condition, including the following items:

- a) Hydrological and geological condition. The UHV AC substation should be located in a stable geological region, and the hydrological and geological influence should be considered.
- b) Meteorological condition. The type, parameters, and configuration of the UHV AC equipment and conductor should adapt to the meteorological condition such as temperature, humidity, atmospheric pressure, wind velocity, etc.
- c) Seismic conditions. The UHV AC substation should avoid being located in a seismically unfavourable site, and electrical facilities should meet the anti-seismic requirements.
- d) Altitude. The safety distance, current-carrying capacity of the conductor, and external insulation of the equipment should be corrected according to the altitude on site.

4.6 Environmental impact

The design of the UHV AC substation should reduce the environmental impact, including the following items:

- a) Audible noise. The type, parameters, and configuration of the UHV AC equipment and conductor should limit the audible noise strength on the substation boundary that is specified by local regulations to a permissible value.
- b) Electric and magnetic field. The type, parameters, and configuration of the UHV AC equipment and conductor should limit the electric and magnetic field strength to a permissible value.
- c) Radio Interference. The type, parameters, and configuration of the UHV AC equipment and conductor should limit the radio interference strength to a permissible value.
- d) The influence of the potential rising of the earthing system on the outside of the UHV AC substation. The potential rising of the earthing system should avoid being transferred to the outside of the UHV AC substation through metal pipes or cables in the design of the earthing system.
- e) Carbon footprint. The design should limit the carbon footprint of the substation (e.g. reduce leakage of total volume of SF₆).

4.7 Economy

The design of the UHV AC substation should strike a balance between performance and cost based on the life-cycle cost analysis, on the premise of meeting functional requirements.

5 Bus scheme and feeder connection

5.1 General

The basic purpose of a chosen bus scheme and feeder connection is to facilitate the operational functions of a substation inside an electrical network. In the past, maintainability and accessibility of high-voltage equipment was very important due to the requirements for frequent maintenance. Different types of circuit breaker design and also the different types of operating mechanisms required regular maintenance with short intervals. These requirements meant that various configurations and arrangements of substations were developed to isolate the circuit breaker and current transformer in a complete bay for maintenance while ensuring availability of supply on adjacent equipment. Disconnectors were required to deal with safety requirements and provide physical isolation during long-term maintenance activities.

Because the latest development in switchgear is aimed at ever longer maintenance intervals, the importance of maintainability in the design of switchgear has changed. At the same time, today's society is getting more and more dependent on electric power supply for all its functions. This results in less tolerance towards quality of power supply issues and black-outs, which will require designers to put more emphasis on high security (i.e. fault tolerance) and availability requirements for substations, especially for UHV AC transmission systems as backbone system.

Summarizing, bus scheme and feeder connection of high voltage substations are strongly influenced by many factors such as operational requirements, security standards, availability, and maintainability, the need for sectionalizing, control and protection systems and regulations. The bus scheme and feeder connection of a UHV AC substation should be determined after comparison of technology and economy according to the importance of the substation in transmission system considering the factors mentioned above.

The configuration for a particular substation depends on its position within a network and also its relative importance.

Because of the generally high importance of UHV AC substations, the following configurations are preferred:

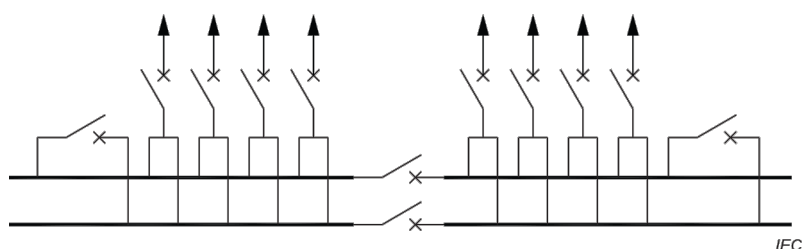
- double busbar (DB);
- one-and-a-half circuit breaker (OHCB);
- two-circuit breaker (2CB).

5.2 Scheme at high-voltage side of main transformer

The DB with a coupler bay as shown in Figure 2 is a substation in which the lines and transformers are connected to either of the two busbars by means of selector disconnectors.

The DB is particularly suitable for highly interconnected power networks in which switching flexibility is important and multiple supply routes are available. The coupler circuit breaker allows the possibility of keeping half of the station in service following a fault on the busbar, a busbar disconnector or any feeder circuit breaker. The configuration provides flexibility by allowing each circuit to be connected to either of the two busbars. It is also possible to move circuits from one busbar to the other while they are energized. Additional flexibility can be provided by adding sectionalised disconnectors into each busbar.

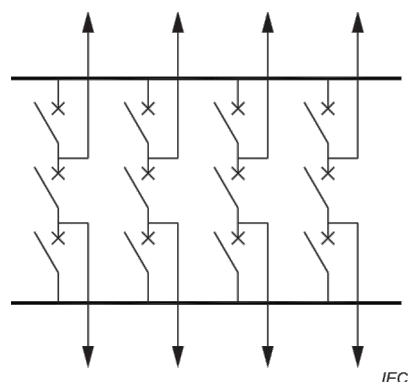
Using a bus section connection arrangement, a higher reliability is provided. This arrangement has the same characteristics and functionality of DB but it is recommended for use when there is a requirement to keep a high number of circuits in service during maintenance or repair of the circuit breaker or the busbar disconnectors.



NOTE The single line diagram only shows circuit breaker and busbar.

Figure 2 – Double busbar (DB) with or without bus section connection

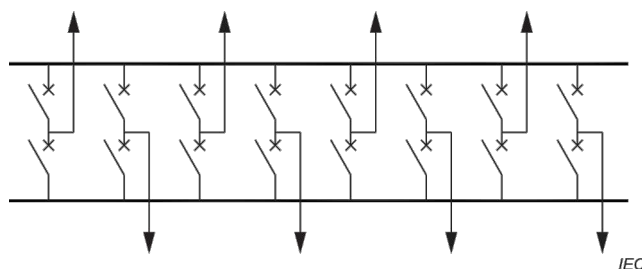
The OHCB as shown in Figure 3 is a double busbar substation where, for two circuits, three circuit-breakers are connected in series between the two busbars, the circuits being connected on each side of the central circuit-breaker. The OHCB is particularly suitable for substations handling large amounts of power, such as those associated with generating stations, and for networks which comprise mainly radial circuits with few mesh connections. The OHCB does not have any separate bus-coupler circuits. Each circuit breaker connecting is acting as a bus-coupler. The design of the bus zone and circuit breaker failure protection systems is simpler than the multiple busbar configurations with selector disconnectors as the systems do not need to select which circuit breakers to trip in response to a busbar fault or a circuit breaker fail situation.



NOTE The single line diagram only shows circuit breaker and busbar.

Figure 3 – One-and-a-half circuit breaker (OHCB)

The 2CB as shown in Figure 4 is a double busbar substation where the selectors are circuit-breakers. The 2CB is recommended for substations where the security of supply is particularly important. In addition, bus section connection in 2CB may strengthen the reliability of the substation. The 2CB is also more flexible than the OHCB. The configuration provides flexibility by allowing each circuit to be connected to either of the two busbars. It is also possible to move circuits from one busbar to the other while they are energized.



NOTE The single line diagram only shows circuit breaker and busbar.

Figure 4 – Two-circuit breaker (2CB)

5.3 Scheme at intermedium-voltage side of main transformer

The following configurations are preferred at the intermedium voltage side of main transformer:

- double busbar (DB);
- one-and-a-half circuit breaker (OHCB);
- two circuit breaker (2CB).

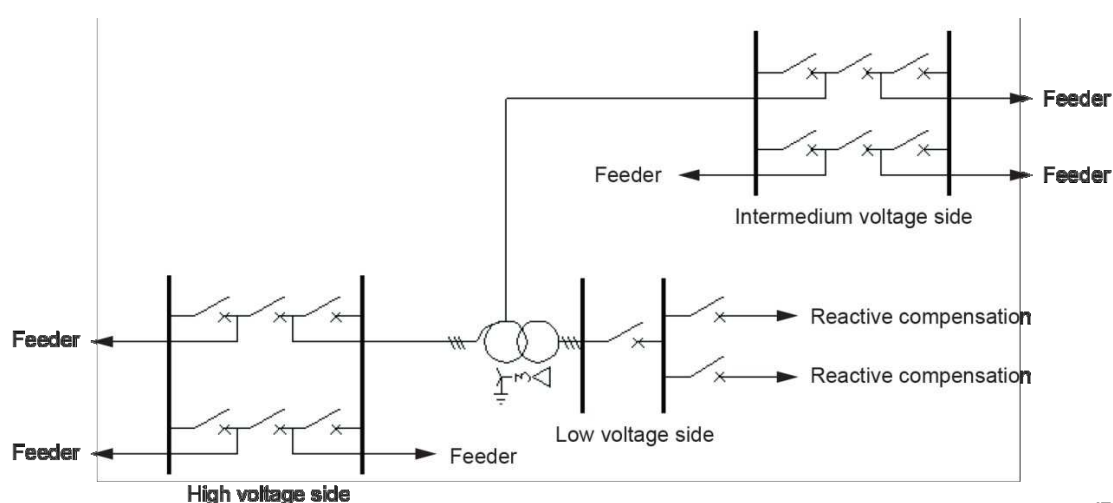
5.4 Scheme at low-voltage side of main transformer

The low-voltage side of main transformer and its rated voltage should be determined according to the following:

- system needs;
- transformer design and manufacturing;
- load configuration (including reactive compensation, feeders, etc.) at the low-voltage side of the main transformer.

If the low-voltage side of the main transformer will be connected to one or more feeders, single busbar layout with bus section connection may be employed. If the low-voltage side of the main transformer is connected to reactive compensation equipment only, the main transformer together with the busbar and connected equipment may be laid out as separate units.

According to load allocation at low voltage side of the main transformer and application of switchgear, a set or more of busbars can be installed at the low voltage side of each bank of transformers. To improve the power transmission reliability of the high-voltage and intermedium-voltage sides of the main transformer and reduce the influence of the low voltage side of the main transformer failure on the operation of the main transformer, each set of busbars may be connected with the low voltage side of the main transformer by a main circuit breaker. Figure 5 shows a typical diagram of a bus scheme and feeder connection.



IEC

NOTE The single line diagram only shows circuit breaker, busbar and transformer.

Figure 5 – Example diagram of a bus scheme and feeder connection

5.5 System neutral earthing mode of a UHV AC substation

A solid-earthed neutral system will result in a very high single-phase short circuit current and require immediate cut-off of lines or equipment, which will increase the operation times of the circuit breaker and reduce the power supply continuity. However, having reduced overvoltage and insulation level, it can lower the equipment cost, especially in HV, EHV and UHV grids, and produce significant economic benefits. When it is required to limit the short circuit current of a system, low impedance earthing can be employed.

The unearthed neutral mode allows running with a fault in cases of single-phase earthing and offers a good continuity of the power supply, in which case the earthing current is limited to the capacitive current of lines and equipment. But owing to the overvoltage, it requires a fairly high level of insulation.

High-voltage side and intermedium-voltage side of a main transformer in the UHV AC substation apply neutral solid-earthed mode or low-impedance earthing mode.

The system neutral earthing mode of the low voltage side of the main transformer should be determined according to the overvoltage, insulation level, power supply reliability and continuity, short circuit current, and protection configuration.

6 Selection of equipment and conductors

6.1 General

UHV AC equipment and conductors should be selected considering the system requirements and environmental conditions.

6.1.1 Voltage

See IEC 60038:2009, Table 5.

6.1.2 Rated current

See IEC 60059:1999 and IEC 60059:1999/AMD1:2009, Clause 3.

6.1.3 Rated frequency

See IEC 60196:2009, Clause 3.

6.2 Basic requirements

6.2.1 Electrical requirements

UHV AC equipment and conductors should be selected considering the following electrical requirement.

a) Short-circuit current

The equipment should be designed and installed to reliably withstand the thermal and mechanical effect caused by the short-circuit current in all kinds of short circuits, e.g.:

- three-phase;
- phase-to-phase;
- phase-to-earth;
- double phase-to-earth.

The peak factor for the rated peak withstand current is related to the DC time constant. The peak factor is higher in the case of UHV AC transmission systems owing to larger DC time constants.

IEC 60909 (all parts) and IEC 60865 (all parts) give the calculation method for the short-circuit current of the three-phase AC system and the effect of the short-circuit current respectively.

b) Overvoltage

The equipment should withstand the switching overvoltage and lightning overvoltage with insulation coordination and overvoltage protection. See Table 3 of IEC 60071-1:2006 and IEC 60071-1:2006/AMD1:2010 for values of rated insulation levels.

c) Electromagnetic interference

Radio interference with a frequency over 10 kHz mainly comes from the current, voltage mutation and corona discharge of the equipment or conductors. It can affect or impair the normal receiving of electromagnetic signal and operation of electrical and electronic equipment. Therefore, the electromagnetic fields generated by the equipment and conductors should be within an acceptable range, and the radio interference caused by the electromagnetic field should not exceed the required value. Because of UHV AC transmission systems, the voltage and current are high; therefore, the interference becomes severe in comparison to the other ratings. Under the highest operating phase voltage, the equipment and conductors should produce no visible corona on clear nights.

The limited value of electromagnetic interference and corona of UHV AC substation should not exceed the existing criteria.

NOTE See applicable limits set by the local regulatory authorities for the acceptable levels of electromagnetic field generated by the equipment and conductors and the maximum allowable radio interference.

6.2.2 Mechanical requirements

The mechanical strength of bushing and post insulator of equipment should be designed considering the effect of wind, ice and earthquake, electrodynamic action of short circuit current, as well as the combination of various loads. Annex A (see Table A.1) shows an example of load combination selection of UHV AC equipment. The safety factor of mechanical strength of bushing and post insulator should also be checked under the long-term and short-term load. The effect of wind and ice should be considered as a long-term load. The effect of earthquake and electrodynamic action should be considered as a short-term load.

When the equipment bushing is connected to bundled conductors, the effect of the conductor rigidity should be taken into consideration.

6.2.3 Environmental conditions

UHV AC equipment and conductors are typically located outdoors. They should be selected considering the influence of the following environmental conditions:

a) Ambient temperature

The equipment and conductors should be able to operate under the specified highest and lowest ambient temperatures. In case ambient temperature is out of the range of specified values, the allowable current-carrying capacity should be subject to agreement between manufactures and users.

The specified value of the highest and lowest ambient temperature refers to Subclause 4.4.2.2 of IEC 61936-1:2010.

b) Solar radiation intensity

As the equipment will experience additional temperature rise under solar radiation, it should be able to withstand the highest solar radiation intensity at midday on sunny days specified if necessary during operation.

Solar radiation level refers to IEC 60721-2-4.

c) Wind velocity

The average maximum wind velocity for many years can be commonly determined according to the observational date of the 10 min mean wind velocity at the 10 m height from the open, plane terrain in locality, through the probability statistics, meanwhile, considering the relative air density. The design wind velocity should be corrected based on the actual installation height. For areas frequently hit by tropical cyclones (typhoon or hurricane) and strong wind, in addition to the purchase of customized equipment, effective protection measures should also be designed, such as lowering the installation height and enhancing the foundation.

d) Altitude

When the altitude is not exceeding 1 000 m, the external insulation of equipment requires no correction. When the altitude is above 1 000 m, correction is required as specified in IEC 60071-2.

e) Ice and snow

Measures such as increasing the effective creepage distance should be taken to prevent insulation flashover in areas exposed to heavy snow and ice. The ice-breaking capacity of the disconnector should satisfy the maximum ice coating thickness of the installation site.

f) Humidity

The equipment selected should satisfy the requirement for the local maximum relative humidity. The specified value of humidity refers to IEC 61936-1.

To withstand the effects of high humidity and condensation, a UHV AC equipment design for such conditions should be adopted. Condensation may be prevented by special design

of the building or housing, by suitable ventilation and heating of the station or by use of a dehumidifying equipment.

g) Pollution

The damage caused by pollutants to electrical equipment in the polluted area depends on the conductivity, water absorption, adhesion, quantity and specific weight of pollutants, and the distance and meteorological conditions. See IEC TS 60815 (all parts) for classification of pollution and creepage distances of equipment under different conditions of pollution. Anti-pollution measures under serious pollution include increasing the effective creepage distance of external insulation, selecting electric porcelain with alternating sheds, large inclinations composite insulators with hydrophobic behaviour or other structures beneficial for pollution prevention.

h) Audible noise

The audible noise at the substation boundary should meet the local environmental protection regulations. See Annex B (Tables B.1, B.2, B.3 and B.4) for specification of UHV AC equipment and conductor.

If the equipment noise exceeds the limit, the following measures can be taken:

- use low-noise equipment;
- install a sound insulation barrier around the equipment;
- install a sound insulation barrier on substation walls;
- install a box-in sound insulation device on the equipment.

i) Earthquake

The earthquake mainly affects the equipment by seismic wave frequency and vibration acceleration. UHV AC equipment is higher, heavier and larger than EHV equipment. The natural frequency of equipment is typically close to that of the earthquake. Accordingly, measures should be taken to avoid resonant frequency and appropriately increasing the damping ratio of the equipment.

In verification of seismic performance for bushings and post insulators, the dynamic magnification factor of the whole structure, the apparatus, the ground conditions including foundation, the support, terminals (such as transformer bushing terminals) and connection fittings should be considered.

NOTE The ratio of vertical ground acceleration to horizontal acceleration is taken as 50 % in IEC 62271-207, 80 % in IEEE 693.

6.3 Transformer

The selection of UHV transformer should be taken into consideration as specified in IEC 60076 (all parts). And regarding higher voltage and capacity than EHV, some aspects such as the type, transportation and cooling mode of the UHV transformer should be given consideration.

Due to the high voltage and large capacity required for UHV AC substations, single phase oil-immersed transformers are recommended considering the transformer's manufacturing difficulty, reliability requirements, and transportation.

The autotransformer is recommended for main UHV transformers. The capacity of common winding in autotransformers should be checked according to the capacity of the system and reactive compensation configuration on the tertiary winding side of autotransformer.

Depending on the rated capacity, the ability of manufacturing and transportation, two-leg or three-leg main transformers may be employed.

As a UHV transformer is heavy in weight and large in size, it is necessary to take the transportation conditions into account. For transportation, the main UHV transformer may be disassembled and reassembled on site.

The main UHV transformer may use the oil forced air forced (OFAF) or oil directed air forced (ODAF) cooling mode to increase the cooling efficiency and ensure that the temperature of the transformer oil and winding meets the requirements.

The voltage of the transformer is regulated by using the tap changer to change the operation voltage ratio. The mode and range of voltage regulation are determined based on the voltage fluctuation range of the system and the requirement of system operation. If the system has a great voltage fluctuation and requires an on-load tap changer, the on-load regulation should be employed; otherwise, the off-circuit tap-changer should be employed.

The voltage regulation winding of the autotransformer may be installed at the end of the HV side winding, the end of the MV side winding or the neutral side. According to the manufacturing ability of the UHV transformer and tap, the neutral side regulation mode may be used.

If the UHV transformer employs the neutral voltage regulation, voltage deflection will occur on the LV side winding. When the voltage on HV side rises, it is required to reduce the number of common windings to stabilize the MV side voltage, which will raise the electrical potential of each winding, further saturate the iron core and increase the LV side voltage accordingly. In the case that the tertiary winding is connected with parallel compensation devices, the voltage change will be further raised, and the operation performance will deteriorate. Therefore, tertiary compensation windings should be added in the main transformer to control the voltage fluctuation range on the LV side.

6.4 UHV shunt reactor and neutral-earthing reactor

The selection of the UHV reactor should be taken into consideration as specified in IEC 60076 (all parts). Some aspects such as type, transportation and cooling mode of UHV shunt reactor should be given consideration.

As the capacity of UHV reactor is relatively large, the impact of electrodynamic force to the neutral earthing reactor during open-phase operation should be considered. The parameters of the neutral earthing reactor should be determined based on the requirements of secondary arc quenching acceleration and resonance overvoltage suppression combining of the power system operation. The insulation of the shunt reactor and neutral earthing reactor should be determined through the system overvoltage calculation.

Typically, the single-phase oil immersed iron core UHV reactor is recommended.

Considering the high rated capacity or transportation, the UHV shunt reactor can be of one leg or two leg type, and oil natural air natural (ONAN) or oil natural air forced (ONAF) should be employed for the UHV shunt reactor.

6.5 UHV switchgear

The developments in the design of high voltage devices and new switchgear components using different design principles with higher reliability or integrated functions mean that the reliable and efficient circuit configurations of the past may not be necessary and may result in onerous life-cycle-cost requirements for utilities. From a product point of view, new high voltage switchgear components are being developed based either on air-insulated technology (AIS) or on gas insulated technology (GIS) or on a combination of both, the so-called Mixed Technology Switchgear (MTS). In addition to the investment costs, life-cycle costs of high voltage substations should be considered for the selection of the technology.

In CIGRE TB 570, an evaluation of the applicability of the various characteristics to the three types of technologies AIS, MTS and GIS is given.

In recent years, environmental requirement aspects have become more relevant and important, and have to be considered when planning UHV AC substations. However, minimizing the impact on the environment of materials and practices used in a substation needs to be balanced with the requirement for equipment to be able to withstand severe environmental and climatic conditions.

6.6 UHV circuit breaker

Circuit breakers are used to connect or disconnect bays from a busbar and/or to connect different busbars or busbar sections. They are the most important element of high-voltage equipment, because they are able to control the power flow in the network and to break fault currents, i.e., to disconnect a faulty part of the network. Because of this important function, the reliability of a circuit breaker needs to be high. The circuit breaker type and design should be selected considering the actual project conditions and equipment manufacture and supply condition, as well as national policies for related industries, and technical/economical comparison. The technical requirement for circuit breakers should be determined after systematic study considering the characteristics of UHV AC transmission systems, based on IEC 62271-100.

UHV circuit breakers have more than one interrupter unit (typically 2 to 4) connected in series. The closing resistor can be positioned in parallel or in series to the interrupter units.

To suppress switching overvoltages, in addition to the surge arresters, pre-insertion closing and sometimes also opening resistors or point-on-wave switching could be applied. The effects of closing resistors are well-known and closing resistors are applied to most of the UHV AC transmission systems. The opening resistor can reduce the switching overvoltage and also have a considerable effect to reduce the amplitude of TRV oscillations. The effect has to be proven by simulation studies.

The values of closing and opening resistors should be taking into account the overvoltages generated by making and/or breaking operations and the insulation design of the transmission systems. The resistor values depend on the system aspects such as network configuration, and to which extent the switching surges are to be controlled.

The insertion time for the closing and/or opening resistors should be determined taking into account the switching overvoltage and insulation design of the transmission system. The insertion time of opening resistors should be set to be longer than the maximum arcing time of the main contacts and they should be inserted for a fixed period, typically around 30 ms, after opening of the main contacts.

6.7 UHV disconnecter

A disconnecter is a mechanical switching device which provides, in the open position, an isolating distance suitable to meet specified electrical withstand requirements. It is capable of opening and closing a circuit when either negligible current is broken or made, or when no significant change in the voltage across the terminals of each of the poles of the disconnecter occurs. It is also capable of carrying currents under normal circuit conditions and carrying for a specified time currents under abnormal conditions such as those of short-circuit. Disconnectors are important to ensure adequate isolation safety conditions inside a substation, particularly during work or maintenance.

Disconnectors in multiple busbar substations need the capability to also switch commutating currents. Commutating currents appear when the load is switched from one busbar to the other due to the voltage drop over the current path. The voltage values are defined in IEC 62271-102.

During switching of disconnectors in gas-insulated switchgear a varying number of pre-strikes and re-strikes occur. Due to the very short duration of the voltage collapse of a few nanoseconds at the switching gap, travelling surges are generated in the GIS. The multiple refractions and reflections of these surges at impedance discontinuities within the enclosures create complex waveforms (very fast front overvoltages – VFFO).

VFFO in GIS are of greater concern at the highest rated voltages, for which the ratio of the lightning impulse withstand voltage (LIWV) to the system voltage is lower. Because the generated transients depend strongly on the specific configuration and on the superposition of travelling waves, it is not possible to give generally admitted values, valid for each case. An accurate simulation for each substation, especially in the UHV range, is necessary for the insulation coordination as a basis for the decision making about possible countermeasures. If the required VFFO withstand voltage is higher than LIWV, it is necessary to define measures reducing the risk of failures. There are two possibilities: an increase of the LIWV connected with additional costs or a mitigation of VFFO. The first choice is easy to realize, but cost-intensive. Nevertheless, in some cases, this solution has advantages. The second choice aims for mitigation of amplitudes of VFFO and finally for a reduction of the effect of VFFO on the equipment.

The damping of VFFO by integration of a damping resistor is a well proven technology. Generally, the mitigation effect of the damping resistor depends on the value of the resistance. Consequently, the resistance of a damping resistor could be chosen and defined according to the maximum calculated VFFO and the required mitigation effect. The damping resistor has to withstand the dielectric stress during striking. A flashover across the resistor may lead to high VFFO comparable to a disconnector without damping resistor and has to be avoided. A higher resistance value leads to a higher voltage stress across the damping resistor and can reach values in the range of 2 p.u. The absorption energy strongly depends on the load side capacitance and the voltage across the disconnector. The required capacity of thermal energy absorption for the resistor could be calculated by summing up all close-open operations containing a high number of strikes. Mostly the thermal absorption capability is defined to withstand the thermal stress for one close-open operation. The possibility to operate more than one close-open operation within some minutes which corresponds to the thermal time constant of the damping resistor is estimated to be very short.

The application of a slow-acting disconnector provides significant reduction of the VFFO, because of lower trapped charge voltages.

6.8 UHV earthing switch for maintenance

A mechanical switching device for earthing parts of a circuit, capable of withstanding for a specified time currents under abnormal conditions such as those of short circuit, but not required to carry current under normal conditions of the circuit. An earthing switch may have short-circuit making capacity.

Earthing switch for maintenance should earth de-energised parts so that work should be performed safely. As GIS is enclosed and the primary conductor is not accessible, each section should be provided with an earthing switch, which means that the number of earthing switches usually is higher than in AIS. The earthing switch in the incoming and outgoing feeders in a GIS normally is designed to be capable of closing onto, and withstanding for a specified time, the short-circuit current, a so-called high-speed earthing switch.

In the case of multiple configurations of overhead transmission lines, current may circulate in de-energized and earthed lines as a result of capacitive and inductive coupling with adjacent energized lines. Earthing switches applied to earth these lines should therefore be capable of switching induced currents.

6.9 High-speed earthing switch

In a UHV AC transmission system, the secondary arc does not extinguish in a short time for high induced voltages because the operating voltage level of UHV is very high and the charging capacity per unit length is also comparatively high. Therefore, a high-speed reclosing scheme should be employed to suppress the secondary arc with a four-legged reactor or high speed earthing switch (HSES).

HSES is often adopted at both ends of a relatively short transmission line. Alternatively, one HSES may be installed at a single end of an extremely short transmission line.

For double circuit systems with multi-phase auto-reclosing scheme, where four-legged reactors are not suitable, application of HSES is a useful and important means for the purpose of secondary arc extinction.

Interlocking is required between HSES and circuit breaker to avoid incorrect operation. Table A.2 of IEC 62271-112:2013 shows a comparison of a four-legged reactor and HSES with the following addition as shown in Table 1.

Table 1 – Comparison of a four-legged reactor and HSES

Equipment	Four-legged reactor		HSES
Connection	Direct connection with GIS	Line entrance side	Between CB and line side DS
Required equipment	Four-legged reactor, (Surge arrester with low protective level) Disconnecter, Earthing switch.	Four-legged reactor, Surge arrester with low protective level	HSES

6.10 UHV current transformer

The selection of a UHV current transformer should be taken into consideration as specified in IEC 60044 (all parts).

Compared with the EHV system, the specification of a UHV current transformer is more strictly for the transient characteristics along with the increase of the DC component damping time constant of the short-circuit current. Requirements should be specified regarding the issue in the current transformer selection.

6.11 UHV voltage transformer

The selection of a UHV voltage transformer should be taken into consideration as specified in IEC 60044 (all parts).

In the case of being installed inside the GIS or MTS switchgear, the SF₆ gas insulated transformer with electromagnetic type or capacitive type can be employed, and in the case of choosing post type, the oil-immersed or SF₆ gas insulated transformer with capacitive type is typically employed.

The type of post capacitive voltage transformer is classified to stack-together structure and stack-separate structure. Selecting the suitable type for UHV capacitive voltage transformer mentioned above, the following aspects besides those for EHV capacitive voltage transformer should be considered:

NOTE Stack-together structure means that a capacitive voltage divider and an electromagnetic unit are placed in one cabinet. Stack-separate structure means that the capacitive voltage divider and the electromagnetic unit aren't placed in one cabinet.

- Low probability that the bottom capacitor of the capacitive voltage divider is affected by the operation temperature rise of the electromagnetic unit, which affects the measuring accuracy;
- Low probability of flashover discharge that occurs between the HV conductors led from capacitive voltage divider and the internal components or shell of electromagnetic unit;
- The convenience of separating the capacitive voltage divider from the intermediate transformer for commissioning or testing.

6.12 UHV surge arrester

The selection of UHV surge arrester refers to IEC 60099-4 and IEC 60099-5.

The surge arrester with low protective level should be used both on the line side and transformer side of UHV switchgear busbar. The surge arrester configuration for UHV switchgear busbar and shunt reactor circuit should be determined based on lightning overvoltage calculation or simulation test taking into account operation mode, shield angle of incoming transmission line section and other factors. The layout location and parameters of the surge arrester should meet the requirements of insulation coordination. As the duration of power frequency overvoltage can be limited, the permissive capability of arrester for power frequency overvoltage should be confirmed during the design stage. See Table 3 of IEC 60071-1:2006 for values of rated insulation levels.

6.13 Reactive power compensation device for low voltage side of UHV transformer

Reactive power compensation devices for tertiary winding of UHV transformer include the shunt capacitor and shunt reactor.

The selection of the shunt capacitor and shunt reactor should be taken into consideration as specified in IEC 60871 and IEC 60076 (all parts) respectively.

The sub-bank capacity of shunt capacitor or shunt reactor should be accommodated with the current capacity of the circuit breaker and should meet the requirement of busbar voltage fluctuation as breaker operation.

Also, the ratio of series reactance to parallel reactance should consider the effects of background harmonics.

The reactor can be dry type or oil-immersed type. If the dry type reactor is used and the single bank of shunt capacitor has a large capacity, the structure of windings in series can be used considering limitation of manufacturing ability and transportation. As all magnetic lines in the dry reactor winding will develop into circuits through air, the anti-magnetic space around reactor should be considered in case it will result in serious electromagnetic induction heating or electrodynamic force in the adjacent conductors (including the earthing conductor). All closed loop connection should be avoided at the magnetic conductive net, fence, support, foundation reinforcement, earthing conductor and secondary wiring.

6.14 UHV bushing

Generally, the UHV bushing is used to connect UHV equipment (main transformer, UHV reactor, UHV circuit breaker and UHV GIS) to overhead lines or UHV AC equipment.

The selection of UHV bushing should be taken into consideration as specified in IEC 60137. In terms of insulation structure, the UHV bushing is classified to the oil-immersed paper capacitance graded bushing and the gas-insulated bushing. The oil-immersed paper capacitance graded bushing should be used for the main transformer and UHV reactor, and the gas-insulated bushing should be used for UHV circuit breaker and UHV GIS.

In terms of material of external insulation, the UHV bushing is classified to porcelain bushing and composite bushing. The selection of porcelain bushing or composite bushing should be based on the pollution level, seismic intensity and other aspects of the substation.

6.15 UHV insulator

The selection of the UHV support insulator should be taken into consideration as specified in IEC 60168 for porcelain and IEC 62772 and IEC 62231 for composite insulators.

The selection of the UHV suspension insulator should be taken into consideration as specified in IEC TS 60815 (all parts).

The suspension insulator string of UHV AC substation should take a V-type string for lowering the gantry of UHV switchgear and reducing the wind-deviation impact of conductors. Also, if V-type string is taken, the deformation of long strings should take into account as selected the apparatus of grading ring and shield ring.

6.16 UHV conductor

6.16.1 General

UHV AC substation conductors are mainly used to connect the electrical device and transmit electrical power. Due to the high current flowing in the busbar, an appropriate conductor should be selected.

The design of the conductor and fittings of the UHV electrical equipment should consider the corona effect on any condition, especially on rainy days.

The design of the conductor of the UHV electrical equipment should consider the influence of the earthquake, calculate the force of the conductor, and select the flexible conductor according to the seismic intensity.

6.16.2 Conductor type

The types of UHV AC substation conductors should be determined according to their current-carrying capacity, thermal stability, mechanical characteristics, economy, site conditions, switchgear characteristics and so on. The conductors should be easy for installation and well adaptable to environmental conditions (high seismic performance, for instance).

In general, tabular conductors, flexible conductors and gas-insulated metal-enclosed conductors may be used in UHV AC substations. The advantages and disadvantages of conductors are shown as Table 2.

Table 2 – Comparison of conductors

Conductor type	Advantages	Disadvantages
Tubular busbars	The conductors have large current-carrying capacity, high corona inception voltage, good structural strength, simple installation fittings, convenient construction and wide adaptability.	The conductors have large stiffness, make the fittings and equipment have great stress during vibration.
Flexible conductors	The conductors have good flexibility, good anti-vibration performance, low cost, and low connection stress.	The conductors have larger phase-to-earth and phase-to-phase clearance is required due to their susceptibility to wind deflection and electromotive force, and higher structural strength is required due to their larger tensile force. In addition, multi-split flexible conductors are required for UHV AC substations because of high voltage level and corona, and are more complex in installation and construction.
Gas-insulated metal-enclosed conductors	The gas-insulated conductors are compact and have high corona inception voltage, no radio interference and electromagnetic issues.	Use of gas-insulated conductors should be decided based on technical and economic aspects.

6.16.3 Selection of current-carrying capacity (cross-section)

Current-carrying capacity should be appropriate to ensure the safe and stable operation of the conductor and equipment, and it should be determined according to the maximum operating current, short-circuit current, heat elimination conditions, etc. of the circuit. For outdoor bare conductors, the ambient temperature, solar radiation, wind speed, etc. should also be taken into account. The maximum continuously operating current should be seen as the operating current, and the possible maximum short-circuit current under various fault conditions as the short-circuit current. Meanwhile, thermal equilibrium and thermal stability should be considered.

6.16.4 Corona and radio interference

Special attention should be paid to the interference of corona and radio in the selection of outdoor bare conductors for UHV AC substations. The very high electric field on the surface of a conductor easily makes the surrounding air ionized and thus forms corona, which causes a corona discharge, increases the losses and brings radio interference and audible noise.

Since the mechanism of corona formation is extremely complex and related to many factors such as conductor shape, outside diameter, surface roughness, phase-ground distance, phase-phase distance, voltage level, contamination, weather conditions and so on, it is very hard to predict accurately whether corona will be formed on the conductor surface by means of any formula.

The same approach may be used in EHV substation about corona inception voltage measurement of bundle conductor in UHV AC substation. Normally, in order to reduce the corona losses, the radio interference level and audible noise, the corona inception voltage of bundle conductor of UHV AC substation should be high enough in a certain level in comparison to the operation voltage. An example of calculation of corona inception voltage of bundle conductor used in UHV AC substation is given in Annex C.

6.16.5 Mechanical strength

The mechanical properties of conductors may be undermined during operation due to their own gravity, wind load, ice load and instantaneous short-circuit forces in case of any failure. The mechanical strength of conductors for UHV AC substations should be checked according to their locations, environmental conditions and types. In order to ensure the safety of equipment, the stress should be checked during the earthquake according to the earthquake intensity demand of sites.

7 Equipment layout

7.1 General requirement of equipment layout

7.1.1 General

General requirements for power equipment layout in substation are introduced in IEC 61936-1. Special care for the design of UHV AC substations is described in 7.1.2.

7.1.2 Optimization of substation layout

To determine the equipment layout, the typical UHV AC substation should be designed considering the equipment function, maintainability and space reduction. The UHV AC substation layout is recommended to be compact and the following general rules are recommended for basic equipment layout:

- equipment is separately installed by unit and voltage, e.g. UHV switchgear unit, main transformer unit;
- share the space of equipment installation and disassembly in case of failure, and optimize equipment installation layout;
- shorten the length of busbars;
- minimize the roads as much as possible and utilize those roads for installation space such as parking and working space of towing cars or temporary location of equipment before installation;
- control the noise at the boundary to fulfil the requirements of local regulation;
- minimize the equipment in order to minimize the overall layout.

7.1.3 Seismic performance

Fundamental mechanical requirements of seismic loads are introduced in Subclause 4.3.9 of IEC 61936-1:2010 and IEC 61936-1:2010/AMD1:2014. As for UHV AC transmission systems, there are cases where UHV AC substations are constructed in areas where the earthquakes are expected. For these cases, a special care is necessary. The details of design for seismic performance are described in Subclause 7.8.

7.1.4 Construction, serviceability and reliability and failure response ability

General requirements for construction, serviceability and reliability and failure response ability of power equipment refer to 4.1.1, 7.1.1, 7.1.4, 7.4.1 and 7.4.2 of IEC 61936-1:2010. UHV AC equipment is normally large and heavy; therefore, UHV AC equipment should be disassembled into separate units for transportation along with consideration of on-site assembly in the UHV AC substation. The number of units assembled in the UHV AC substation should be minimized.

The procedures in case of equipment and/or system failures should be considered at the design stage. Careful consideration of these procedures is imperative in the design of UHV AC substations.

Outage area of substation in case of failure, maintenance or future expansion should be minimized. Examples of recommendations:

- multiple busbar to avoid whole busbar outage, or function of separation of outage area of busbar;
- gas compartment arrangement of depressurization to low pressure and vacuum evacuation for GIS and MTS;
- location of detachable busbar (expansion joint) for disassembly in case of failure;
- location of back-up (stand-by) equipment for replacement in case of necessity.

7.2 Minimum clearances

7.2.1 Normal environmental conditions

See Subclause 4.4.2 of IEC 61936-1:2010.

7.2.2 Minimum clearances in air-voltage range

See Subclause 5.4 and Table 2 of IEC 61936-1:2010 and IEC 61936-1:2010/AMD1:2014.

NOTE The minimum clearances in air given in Table 2 of IEC 61936-1:2010 apply for altitudes up to 1 000 m above sea level. For higher altitudes, see Subclause 4.4.3.2 of IEC 61936-1:2010.

The duration of the wave front is about a few hundred μs , such as the overvoltage in opening/closing transmission lines and earth fault. This switching overvoltage has much influence on the insulation design of towers and is particularly important for UHV AC transmission systems because of the saturation effects of the air insulation distance on the switching impulse strength. The flashover voltage of air-insulated gaps for switching overvoltage has a tendency to saturate. Therefore, an extremely high tower is required for air insulation. On the contrary, to reduce the construction cost of UHV AC transmission system, switching overvoltages can be reduced by adopting circuit breakers with closing and/or opening resistors and surge arresters with low protective levels.

7.3 Electromagnetic environment

7.3.1 Electrostatic induction mitigation design

See Subclause 4.2.7 of IEC 61936-1:2010 and IEC 61936-1:2010/AMD1:2014. The design of an installation should be such as to limit the electric and magnetic fields generated by energized equipment to an acceptable level for exposed people who are performing safety, operational and maintenance work. Further information is available from the International Commission on Non-Ionizing Radiation Protection (ICNIRP).

In a publication provided by the International Commission on Non-Ionizing Radiation Protection (ICNIRP) named "ICNIRP Guidelines for limiting exposure to time-varying electric and magnetic fields (1 Hz to 100 kHz)-2010", guidelines are established for the protection of humans exposed to electric and magnetic fields in the low-frequency range of the electromagnetic spectrum from 1 Hz to 100 kHz. The main objective of this publication is to establish guidelines for limiting exposure to electric and magnetic fields (EMF) that will provide protection against all established adverse health effects. Accordingly, based on this publication, reference levels for occupational exposure to electric fields, which is defined as time-varying electric fields in unperturbed RMS to which a person may be exposed without an adverse effect and with acceptable safety factors, are $5 \times 100/f \text{ kV m}^{-1}$ (f in Hz) for 25 Hz to 300 Hz range, i.e. 10 kV/m for 50 Hz and 8,3 kV/m for 60 Hz respectively. On the other hand, reference levels for general public exposure to electric fields are half of those for occupational exposure.

Therefore, design criteria of ground electric fields generated by electrostatic induction should be less than 10 kV/m at 1,5 m from the ground, considering all the established adverse health effects, especially in the brain. If it is difficult to design less than 10 kV/m, the adoption of

partial shield facilities should also be studied. As for the head region of aerial bushings and lower-part shields, it is necessary to configure those components at a ground height that complies with the above-mentioned permissible electric field.

The following other measures against electrostatic induction should also be considered during the design stage:

- placing any accessible equipment such as local control panels at a lower electric field;
- placing live parts of equipment or incoming leads at a higher location;
- other improvements in shield design to reduce the electric field.

7.3.2 Magnetic induction mitigation design

In accordance with "ICNIRP Guidelines for limiting exposure to time-varying electric and magnetic fields (1 Hz to 100 kHz)-2010", magnetic field strength 1,5 m above the ground should not be greater than 1 000 μ T (50 Hz/60 Hz).

The reasonable measures against magnetic induction are shown as follows:

- secondary equipment such as local control panel may be located as far as possible away from dry-type air-core reactor;
- placing live parts of equipment or incoming leads at a higher location;
- reduce the exposure time on the area around dry-type air-core reactor.

7.3.3 Audible noise mitigation design

See Subclause 4.5.2 of IEC 61936-1:2010.

It is difficult to calculate or estimate the radio noise level because of the irregular shield of shielding rings of equipment in a UHV AC substation. Some effective suggestions to lower the radio noise level based on the engineering experience are provided here.

The shield ring design for bushings, V-shaped suspension insulators, etc. is based on the suppression of the audible noise level in busbars by lowering the corona discharge in an appropriate manner.

It is recommended that during the actual design stage, corona noise levels be below the criteria for the various connection parts between conductors and shield ring for bushing, V-shaped suspension insulator, etc.

The reasonable measures against audible noise are shown as follows:

- increased diameter of end shield (spherical shape) of disconnecting switch;
- enlarged conductor diameter of disconnecting switch;
- enlarged diameter of primary conductor of pull-down leads;
- enlarged conductor diameter for reducing the surface electric field strength;
- using a bundled conductor with more split numbers;
- placing incoming conductor at a higher location.

Audible noise caused by corona discharge under rainy condition should be considered, since the noise level under rainy condition is normally much higher than that under clear weather. An example of conductor design of a UHV AC substation under rainy conditions is given in Annex D.

7.4 Selection of switchgear equipment

For general requirements concerning the selection of power equipment including switchgear, see Subclause 6.1.1 of IEC 61936-1:2010. For UHV AC equipment, there are three types of switchgears: AIS type, MTS type and GIS type.

The technologies used in AIS, MTS and GIS have their own advantages. There is no recommended general solution for all users, the individual conditions of every case having a strong impact on the evaluation and leading to different conclusions. Typical examples of items to be considered in case of selection of AIS, MTS or GIS are as follows and examples of investigation for each item are introduced in Annex E.

- reliability;
- maintenance and operation;
- installation;
- layout;
- circumstance;
- construction cost;
- testing / commissioning.

The principal technologies of AIS, MTS and GIS used in UHV AC substation are shown in Table E.1.

Recently, there are cases to adopt MTS and GIS in a UHV AC substation. In these cases, the following matters should be considered:

- residual charge of main circuit effect against insulators used in MTS and GIS;
- induced surge effect to secondary circuit.

Selection of switchgear type of MTS or GIS is often made from the typical points of view as follows:

- number of live parts (i.e. bushings and busbars);
- anti-seismic performance;
- noise;
- electromagnetic circumstances;
- space of land;
- cost.

7.5 Switchgear Installations layout

7.5.1 General

For general requirements about power equipment installations, see Subclauses 7.1, 7.2 and 7.6 of IEC 61936-1:2010. The layout of switchgear in a UHV AC substation should be determined by function, performance, maintainability, construction and failure response in an integrated manner and not only for installation. As a basic layout, the following recommendations should be considered.

7.5.2 Location arrangement of switchgear

For general requirement of circuit arrangement of power equipment, see Subclause 7.1.1 of IEC 61936-1:2010. As for UHV AC substation, UHV and lower voltage switchgear should be installed in separate locations such as on opposite sides of transformers.

7.5.3 Basic arrangement of surge arresters

For general requirements of design and position of the surge arrester, see Subclause 6.2.5 of IEC 61936-1:2010. The arrangement of surge arresters should be determined through lightning surge and internal overvoltage analyses by varying system conditions including UHV switchgear, transformers and other equipment. The number of surge arresters in the UHV AC substation should be optimized and minimized by analyses. Surge protection and insulation coordination should be considered at the same stage as surge arrester analysis. The basic arrangement of surge arresters is recommended as follows:

- GIS: surge arrester should be located at the outgoing/incoming line, bank and main busbar;
- MTS: surge arrester should be located at the outgoing/incoming line and bank and, at the user's option, located in the main busbar;
- AIS: surge arrester should be located at the line and bank entrance.

7.5.4 Optimal gas-insulated busbar (GIB) layout and length

For general requirements of circuit arrangement for power equipment installations, see Subclause 7.1.1 of IEC 61936-1:2010 and IEC 61936-1:2010/AMD1:2014. UHV AC substation needs adequate large distance between neighbouring gantries for outgoing/incoming transmission lines and transformers. Therefore, location of gantries and transformers should be determined for optimal GIB layout and length.

7.5.5 Utilization of working space for substations

For a general requirement concerning aisles and access areas, see Subclause 7.1.4 of IEC 61936-1:2010.

a) GIS substation

Since UHV apparatus is large and heavy, space for installation of switchgear should be shared with the space for disassembly in case of failures in order to minimize the total area of the substation. For example, one unit of gas-insulated switchgear equipment such as circuit breaker, disconnecter, earthing switch, excluding main busbars and GIB, is recommended to be connected in a linear configuration.

The following items should be considered with the layout of a UHV GIS substation:

- space for equipment needed for installation or disassembly, such as cranes;
- space for vertical or horizontal removal of disassembled equipment;
- space for working, like space for handling of bolts;
- maintenance space, such as space for opening of covers, or manholes;
- space for on-site test.

An example of crane location for installation or disassembly work is shown in Figure 6.

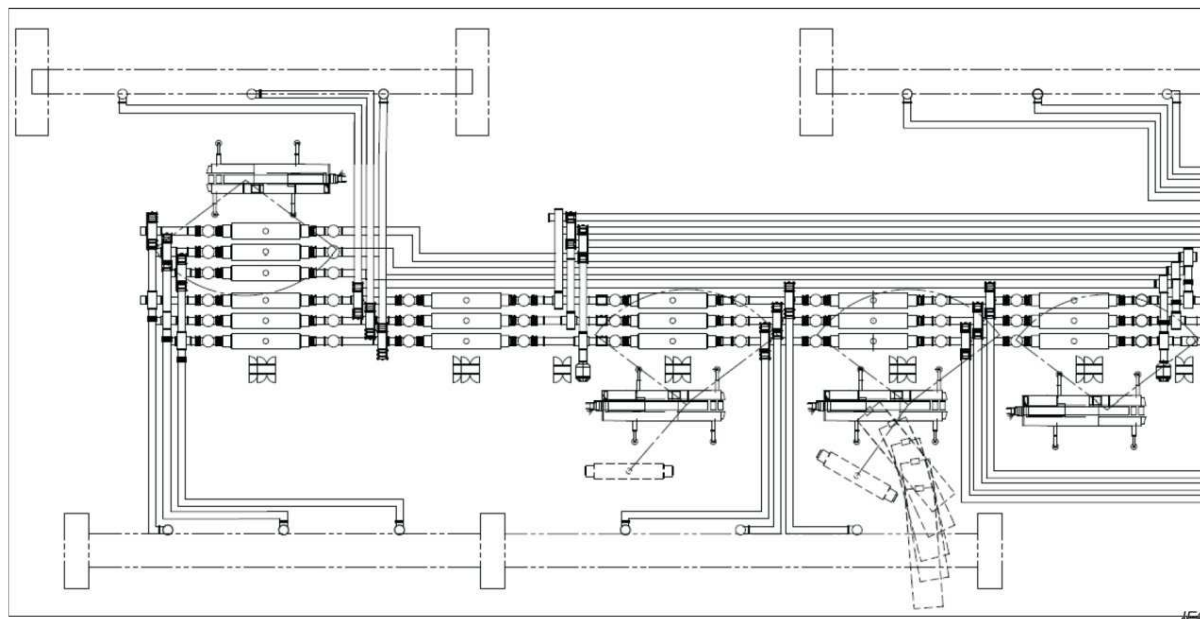


Figure 6 – Typical configuration of UHV gas-insulated switchgear and crane location

b) AIS substation

In case of AIS substations, generally sufficient space is available for installation and removal of any equipment due to the clearance requirement of UHV AC transmission systems. Roads between transformers, circuit breakers and disconnectors may be shared temporarily for assembly and disassembly, testing, or commissioning of equipment, in order to optimize land usage. While considering the layout, space for stationing on-site test equipment such as AC voltage test, impulse test etc. may also be considered.

7.6 Protection against direct lightning strike

For methods of protection against direct lightning strike, see Subclause 8.6 of IEC 61936-1:2010.

a) AIS or MTS substation

Overhead ground wires and/or lightning rods should be installed to shield against lightning in order to prevent a direct lightning strike to the main circuits of the substation.

b) GIS substation

Lightning protection facilities inside the lead-in transmission line of the GIS substation can be omitted for the following reasons:

- the conductive parts are shielded after the entrance of transmission line;
- facilities are not damaged because the energy of the direct lightning strike is extremely small even if it does occur;
- frequency of the direct lightning strike that makes backward flashover occur between GIS enclosures and main circuit conductors is extremely low (less than once per thousand years).

7.7 Earthing systems

7.7.1 General considerations

The earthing systems should be designed on the basis of compliance with IEC TS 60479-1 with the purpose of protecting the entire UHV AC substation.

For the design of earthing systems, see Subclause 10.3 of IEC 61936-1:2010 and IEC 61936-1:2010/AMD1:2014.

Considering that the earth fault current may be large in UHV AC transmission systems, the safety of operators and workers should be assured against the increase of touch and step voltages. The touch and step voltages should both be below a permissible level when an earth fault occurs inside or outside the substation.

When adopting GIS and MTS, the multiple-earthing method should be employed to reduce electromagnetic induced current in secondary circuits.

It is desirable to lay auxiliary earthing mesh under switchgear, whether substation equipment is GIS or MTS, to reduce the electromagnetic induced surge voltage in secondary circuits during an earth fault. There are also cases applied to AIS substation. In addition, the auxiliary earthing mesh material should be resistant to corrosion.

In case of MTS, the earthing current into auxiliary mesh becomes large and the appropriate earthing wire size should be considered. Interphase earthing wire can lead below the ground to realize the circuit connection.

7.7.2 Multiple point earthing method for GIS

Multiple point earthing design is employed for GIS enclosures (see Figure 7). It should be adopted for the following reasons:

- suppression of induced surge from main circuit to secondary circuit;
- earthing is improved by multiplying the number of connections;
- reduction of eddy current heating of the supporting structure due to leakage flux;
- induced current from each enclosure sheath to the ground generated by multi-earthing method with adoption of a phase-to-phase shunt bar placed at the proper distance.

In case that it is difficult to install phase-to-phase shunt bars such as single point earthing parts, the single point earthing method should be employed. In that case, flashover may be avoided by installing insulated flanges between multi-earthing enclosures and bushing parts.

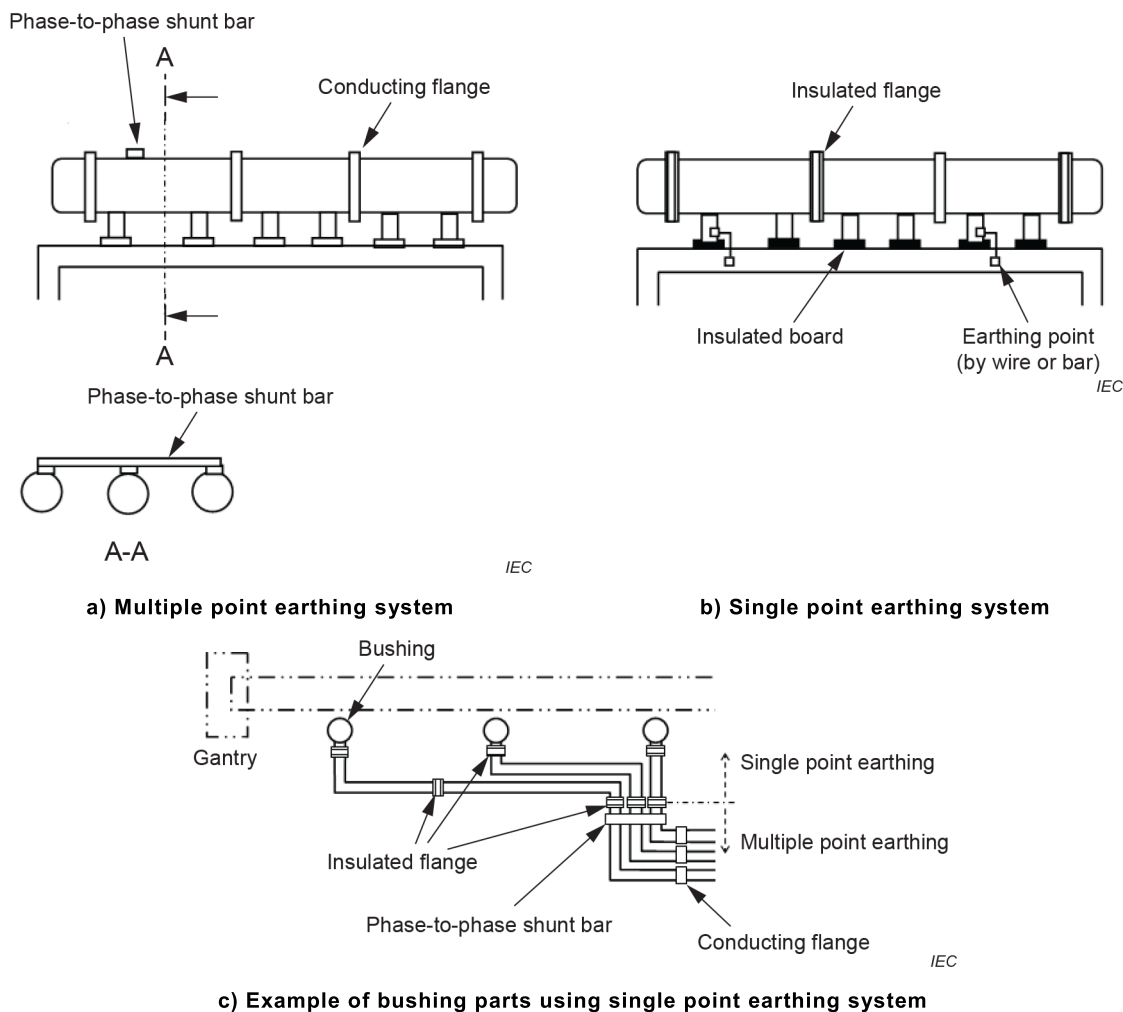


Figure 7 – Earthing methods

7.8 Seismic design

7.8.1 General

Specifications for seismic performance of the equipment depend on the local regulations of each region, country, area, site, condition of basement, and foundation construction.

This means that the seismic criteria, qualification methods and levels, and performance requirements are standardized for a given equipment type.

Therefore, there are some differences in several guides and standards (see Tables F.1 and F.2 of Annex F).

These differences depend on the conditions and damage of past earthquakes in each region and it is very difficult to harmonize these standards.

7.8.2 Basic seismic design

For seismic design of UHV AC substation switchgear, see IEC 62271-207 and IEC TR 62271-300, and for the design of UHV AC substation transformer, see IEEE 693-2005.

For the consideration of seismic loads in seismic design, see Subclause 4.3.9 of IEC 61936-1:2010 and IEC 61936-1:2010/AMD1:2014.

In a region where earthquakes occur frequently, three qualification levels are recommended according to their frequency: low, moderate, and high.

For seismic design, the following flow chart and procedures in Figure 8 are recommended:

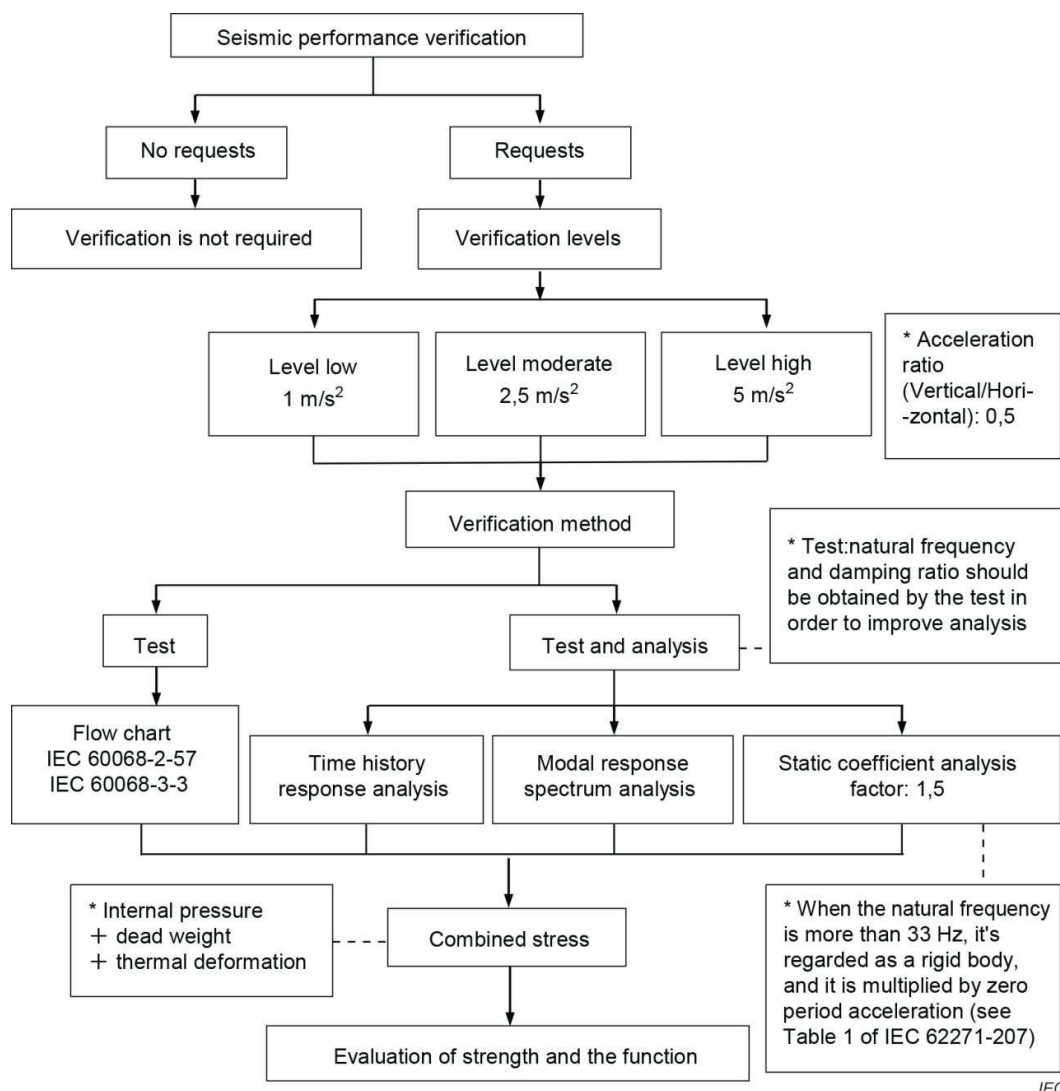


Figure 8 – Flow chart for seismic qualification

- Gas bushing, surge arrester, oil-immersed transformer, VT, PD, MTS, GIS, and AIS exceeding 800 kV should be qualified for seismic performance before they are installed at a location where an earthquake can occur.
- A series of seismic tests should be carried out with a full-scale model whenever possible. However, if it is impossible to conduct the seismic test with a full-scale model due to the size of the actual equipment, seismic analysis is often applied. In this case, mathematical models including a computer-simulation model or sub-set (partial model) which can meaningfully represent the whole system may be used to simulate the seismic performance.

7.8.3 Special seismic performance for UHV AC substation equipment

Seismic performance should be taken into account for large and heavy UHV AC equipment (especially porcelain/composite bushings, porcelain/composite surge arresters, voltage transformers, and supporting insulators).

For seismic considerations of UHV AC substation switchgear, see IEC 62271-207 for GIS and IEC TR 62271-300 for circuit-breakers.

From the point of view of the seismic strength, components made of porcelain materials are considered to be the weakest structures used in UHV AC substations.

a) UHV porcelain/composite bushing

See IEC TS 61463.

b) Porcelain/composite surge arrester

The appropriate dynamic amplification factor (e.g. 1,4 to 2,0) is considered for a porcelain/composite surge arrester mounted on a supporting structure for UHV AC equipment when tested or computed without a supporting structure.

c) Other considerations

The following should be considered in the seismic design of the equipment:

- rocking motion of foundation;
- ensuring sufficient slack of connecting leads;
- split foundations.

8 Control, protection and communication

8.1 General

The function and basic component of the control, protection and communication system of an UHV AC substation is the same as that of an EHV substation.

Because of the importance of the UHV AC substation, the interruption of the substation control system network will affect the safe operation of the substation. The control system network of UHV AC substations should be designed in double-set configuration.

When the UHV transformer is of the split type, that is, provided with the independent regulation transformer and compensation transformer tank (sharing the same tank), if the regulation transformer or compensation transformer has small inter-turn short circuit faults, the UHV transformer longitudinal differential protection designed for the whole transformer is hard to operate. Therefore, it is necessary to provide a separate main protection for the regulation transformer and compensation transformer.

The control, protection and communication system in a UHV AC substation should be configured considering both short-term and long-term needs and have both openness and compatibility.

8.2 Control system

The substation control system should use the networked, distributed and systematic hardware structure conforming to international standards. The computer storage and processing capacity should be able to satisfy the final requirement of a UHV AC substation.

The substation control system consists of two parts, station level and bay level. The station level, as the control, monitoring, measurement and management center of all electrical equipment in the station, serves to manage and control the bay level equipment and communicate with the remote control centre. It consists of a station computer, operator workstation, remote communication equipment, etc. which are connected to the network. The bay level consists of control units classified based on the voltage level and electrical bays. In case of a station level equipment or network failure, bay level equipment should be capable of monitoring and controlling the bay equipment locally and independently. The bay level equipment may be disconnected when the equipment in the bay is under maintenance.

The substation control system should have at least the functions of data acquisition and processing, switchgear and transformer tap control, synchronization for circuit breaker closing, event and alarm record, database management, graphical screen generation and display, measurement trend and energy record, remote communication, etc. The communication protocols of a substation control system should follow IEC 61850 (all parts).

8.3 Relay protection

8.3.1 General

Relay protection of UHV AC equipment generally includes UHV transformer protection, UHV busbar protection, UHV reactor protection, UHV line protection, UHV breaker failure protection, etc.

Relay protection of UHV AC equipment serves to ensure the safe operation of UHV AC equipment and the power grid and to improve power supply reliability of the power grid.

8.3.2 Duplicated configuration of UHV AC equipment relay protection

The configuration of UHV AC equipment relay protection should be considered based on its role and function. In general, UHV AC equipment has a large capacity and a high voltage, and is expensive to purchase and repair. Its safe operation is vital to the system stability as well as the safety and reliability of the substation; besides, because the principle and structure of UHV relay protection are relatively complicated and involve many factors, any UHV AC equipment damage or power loss of a substation due to failure or maloperation of protection will result in huge direct or indirect losses, which imposes higher requirements on the relay protection of the UHV AC equipment.

Therefore, to ensure the safety and continuous operation of UHV AC equipment and minimize the effect of fault and abnormal operation on UHV AC equipment and power systems, it is recommended to provide duplicated relay protections for UHV AC equipment all the time. When one set of protection devices fails, the other set should be able to provide a complete relay protection.

Based on the considerations above, duplicated configuration of UHV transformer protections, UHV busbar protections, UHV reactor protections and UHV line protections should be provided. The UHV breaker failure protection serves as the local back-up of the above protection, while the UHV transformer non-electrical protection and UHV reactor non-electrical protection serve as the supplementation of electrical protection. To simplify the protection configuration, it is recommended to provide only one set of such protections, which should operate on two tripping coils of the circuit breaker at the same time.

8.3.3 UHV transformer protection

UHV transformer protection should consider all kinds of UHV transformer and their operating conditions comprehensively.

a) Purpose of UHV transformer protection

The UHV transformer should be provided with protection against phase-to-earth faults, phase-to-phase faults, winding inter-turn faults, overload, over-excitation, and non-electrical faults.

b) Configuration and function requirements for UHV transformer protection

UHV transformer protection should include the main protection, the backup protection, non-electrical protection and the abnormal operation protection.

The main protection of the UHV transformer should be provided with longitudinal differential protection, which serves to protect the transformer inside, bushing, and outgoing line up to CT.

The backup protection of the UHV transformer is used to protect the overcurrent and neutral overvoltage caused by external earthing and phase-to-phase short circuit and serves as backup protection against faults in adjacent components and the transformer.

The non-electrical protection of the UHV transformer generally covers the gas, low oil level, high temperature, high pressure, cooler fault, etc.

The abnormal operation protection covers the overload protection and over-excitation protection.

c) Configuration and function requirements for regulation transformer and compensation transformer protection

When the UHV transformer is of the split type, that is, provided with the independent regulation transformer and compensation transformer tank (sharing the same tank), theoretically internal faults of the regulation transformer and the compensation transformer should be within the longitudinal differential protection range of the UHV transformer.

But the analysis and results of related dynamic simulation tests of protection devices show that as the regulating transformer and compensator transformer has fewer turns than the main transformer, when the regulation transformer or compensation transformer has inter-turn short circuit faults, especially small inter-turn short circuit faults, the UHV transformer longitudinal differential protection designed for the whole transformer is hard to operate. Therefore, it is necessary to provide separate main protection for the regulation transformer and compensation transformer, that is, differential protection for the regulation transformer and differential protection for the compensation transformer.

Independent non-electrical protection should also be provided for the regulating transformer and compensator transformer, which covers the gas, low oil level, high temperature, high pressure, cooler fault, etc.

8.4 Communication

The communication equipment in the UHV AC substation should be configured according to the local dispatching system and the location of the substation in the grid, and adequate backup equipment should be reserved accordingly. The designation of the communication system of the UHV AC substation should satisfy the requirements of local technical standards.

8.5 Electromagnetic compatibility requirements for control and protection equipment

The UHV AC substation has a fairly complicated electromagnetic environment. The control and protection equipment should have sufficient anti-interference capacity. The main sources of electromagnetic interference are as follows:

- operation of HV disconnector and circuit breaker;
- lightning attack lines, structures or relay rooms;
- power system short-circuit fault;
- power frequency electromagnetic interference of HV lines;
- partial discharge (corona, creeping discharge);
- switch operation of the control and protection circuit;
- LV power supply, such as voltage fluctuation, voltage sag and short-time power failure, power frequency change, harmonic, etc.

It should be noted that the interference immunity concerns all equipment as a whole. Even though a single component, or a combination of components can pass the interference immunity test, coupling of further elements may lead to an increased value or a new mode of electromagnetic interference; eventually, the complete system may fail to meet the requirements of immunity.

Since there are multiple kinds of interference caused by switching surge, operation of HSES, VFFO and so on, the investigation on integrated countermeasures against various surges and overvoltages is desirable.

See IEC 60255-26 for other anti-interference requirements of UHV AC substation control and protection equipment.

9 DC and AC auxiliary power supply system

9.1 General

The function and basic component of a DC and AC power supply system of a UHV AC substation is the same as that of an EHV substation.

Because a UHV AC substation is large in size and contains a large number of control and protection components, a single DC power supply system may not be sufficient in terms of reliability and economy. Several DC power supply systems provided in the substation are recommended, each of which is an independent and complete power supply system located where the loads are relatively concentrated.

The AC/DC auxiliary power supply system should be designed based on the requirements of various loads for voltage fluctuation range and power supply capacity.

9.2 DC power supply system

The substation DC power supply system can employ the centralized configuration or distributed configuration. The centralized configuration means the whole substation has only one DC power supply system. This way is suitable for substations with a small size or concentrated DC loads. The distributed configuration means several DC power supply systems are provided in the substation, each of which is an independent and complete power supply system. This way is suitable for substations with a large size and multiple relay rooms. It is recommended to use distributed DC power supply systems in the UHV AC substation.

The DC power system voltage of a UHV AC substation should be determined appropriately based on electrical equipment type, rated capacity, power supply distance, installation site, etc.

As the main transformer, shunt reactor, UHV line, UHV busbar of UHV AC substation are all provided with duplicated protection and duplicated trip circuits of UHV circuit breaker are also required, the DC power supply system should be provided with two storage batteries for the sake of reliable power supply. The two storage batteries should be connected with two single busbars, between which the electric connection device should be provided. Under normal circumstances, the two storage batteries should run independently with no electrical connection, which helps to improve the power supply reliability of the DC system.

The number and capacity required of cells should be designated based on the DC power system voltage, the DC loads and the time of the power failure occurs at the substation.

It is recommended to provide two sets of charging devices for the two storage batteries of the UHV AC substation DC power supply system. To improve the reliability of a DC power supply system, a third set of charging devices can also be provided, which can charge either of the storage batteries through the electric switching device.

9.3 AC auxiliary power supply system

AC auxiliary power supply system needs to operate the total auxiliary load and play an important role to maintain any control function in the substation. The following should be considered:

- a) Voltage, circuit configuration and control/protection method of the AC auxiliary supplies should be designed based on the bus scheme, selection of equipment, equipment scale of the substation, and reliability to be required.

- b) The capability of load and auxiliary transformers should be designed to accommodate the final load determined during the planning stage.
- c) The AC auxiliary circuit should be designed for economy, operation and maintenance.
- d) An automatic changeover function should be provided in case of loss of power supply. Duplication of power supply is recommended.
- e) Selection of a circuit system should be considered from economy.

9.4 AC uninterruptible power supply (UPS) system

As the host and network equipment of control system employs the duplicated configuration, the UPS host should also employ the duplicated configuration. In the duplicated configuration, two single busbars or a single busbar with bus section connection can be used for connection.

The UPS AC power source of a UHV AC substation should be provided by the auxiliary AC power system, and the DC power source should be provided by the auxiliary DC power system. It is not recommended to provide a special battery for UPS.

10 UHV gantry, support and foundation design

10.1 UHV gantry and support design

10.1.1 General

The structural type and calculation criteria of UHV gantry and support should be determined based on electrical layout, importance of the project and environmental conditions.

- a) The basic design parameters of gantry and support, including design service life and importance factors of structure (importance factors are used to adjust the minimum design loads for UHV structures, depending on the risk to human life, health, and welfare associated with their damage or failure by nature of their occupancy or use) should conform to the latest international standard as well as regulations of the specific country.
- b) According to the importance of UHV AC substation, the limit state design method based on the probability theory can be used in gantry and support calculation.
- c) The gantry should prefer steel pipe or angle steel lattice column and steel pipe or angle steel lattice beam with good stress and aseismic performance. The structural type of the support should be consistent with the gantry. Steel pipe or angle steel lattice support should be preferred.
- d) The allowable deflection of gantry and support in serviceability limit state should meet the requirements of safe operation of the electrical equipment.

10.1.2 Load and combination of loads

Load and combination of loads on UHV gantry and support should consider items as follows:

a) Load classification

Load on the gantry and support can be divided into the following categories:

- Static load: self-weight of structure, lines and fixed equipment on the gantry (including the insulator string, fitting, line trap and so on);
- Dynamic load: wind load, ice load, load from the conductor, temporary load during erection and repair, thermal response, and seismic response.

b) Wind pressure and velocity

The wind pressure and velocity used in gantry and support calculation should be consistent with the important factors and wind sensitivity of structure. The gantry and support should be designed considering the effect of wind vibration and wind vibration coefficient (height factor) should be determined according to the ground surface roughness and height above the ground.

c) Seismic response calculation

The seismic response calculation of gantry and support should be considered as follows:

- Gantry: Seismic response of the gantry should be calculated according to the finite element method. The analysis mode of the seismic response of gantry should be consistent with that of the static force effect; in intensity consideration, axial (tensile) force/bending moment as well as shear force are also necessary;
- Support: The support should be combined with the electrical equipment mounted on the support in calculating the seismic response according to the finite element method.

d) Load combination on gantry and support

Load combination on gantry and support should be determined as follows:

- The design of gantry and support should meet the load-bearing limit and the common operation limit respectively, and their structure should meet the requirements for overall stability and local stability.
- The strength limit state should be determined under three load cases, namely operation, erection, and maintenance. Under operation load case, gale, ice coating and wind, and thermal response should be combined respectively. In serviceability limit state, the erection load case of the gantry can be used as the basis for the calculation of gantry deformation; the quasi-permanent combination value of the wind under the gale load case can be taken as the loading condition for deformation calculation in serviceability limit state of the support.
- The combination of the seismic response should be considered for the gantry and support.
- The load combination factors of lines and lightning wires should conform to the currently valid international standard as well as regulations of specific country.

10.1.3 Detailing requirements

The design of UHV gantry and support should meet following requirements:

- a) Appropriate measures should be taken for use of the continuous gantry structure in Figure 9, considering the thermal effect.
- b) Different sections of the main member in the gantry and support should be connected with the flange or bolt to reduce the welding work at site.
- c) The gantry column should be designed with the ladder, safety cage and other safety devices to facilitate maintenance personnel climbing up and down. The walkways and rail should be provided in the gantry beam for the sake of installation and maintenance.
- d) Effective anti-corrosion measures should be taken according to the different atmospheric corrosion mediums for the gantry and support. Steel structures in common environmental conditions should be protected from corrosion by a hot-dip galvanizing or zinc spray.

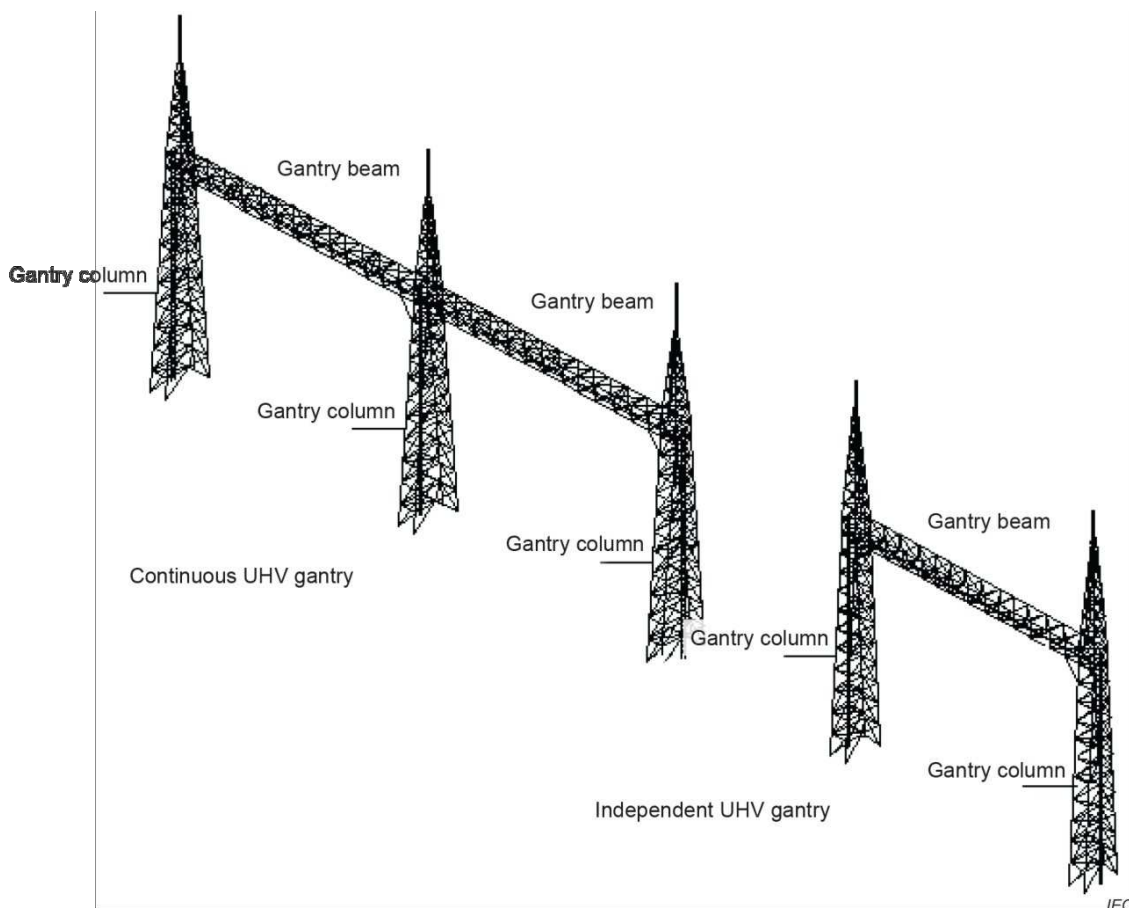


Figure 9 – Example of continuous UHV gantry and independent gantry

10.2 GIS or MTS foundation design

The spread foundation should be installed in the following ground:

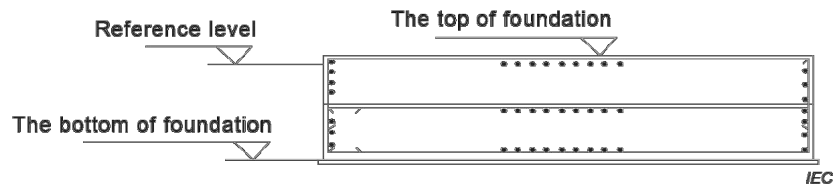
- bearing force of ground can be expected;
- harmful settlements do not occur to a superstructure;
- groundwater level is relatively low.

Meanwhile, when installing in the ground where bearing force of ground cannot be expected, the pile foundation or the spread foundation by ground improvement should be installed.

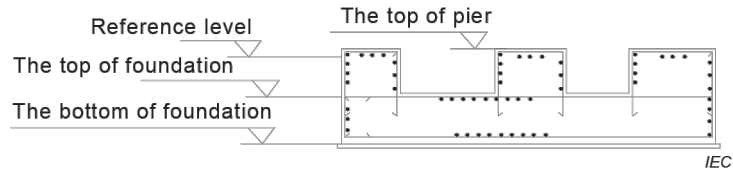
GIS or MTS foundation design should consider items as follows:

- GIS should adopt the unified raft foundation, raft foundation with upper buttress, or unified box foundation. Within the GIS foundation plane, cable trenches, embedded iron parts, and earthing points are densely, complexly and irregularly set, so GIS foundation should use raft foundation with upper buttress preferentially. The examples of GIS foundation forms are as shown in Figure 10.
- When the GIS foundation is too long, efforts should be made together with the equipment manufacturer to set an expansion joint or settlement joint at the corresponding position of the horizontal expansion joint and vertical expansion joint of gas-insulated busbars.
- The temperature cracks generated by hydration heat impact in mass concrete construction should be prevented or controlled. The design should clearly specify the type of foundation concrete and type and quantity of additives; the surface should be provided with anti-crack reinforcement, and the foundation plate should be provided with post-cast belt.

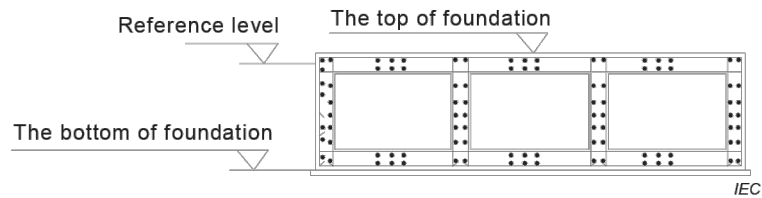
- d) In order to monitor the settlement of the foundation in real time, the settlement observation point may be adopted in a position that is mutually visible.



a) Unified raft foundation



b) Raft foundation with upper buttress



c) Unified box foundation

Figure 10 – GIS foundation forms

Annex A (informative)

Load combination of UHV AC equipment

The following Table A.1 is an example of load combination for UHV AC equipment.

Table A.1 – Example of load combination for UHV AC equipment

Circumstance	Wind velocity	Net weight	Downlead conductor weight	Ice weight	Short-circuit force	Seismic force
Normal	Wind speed with ice	√	√	√		
	Maximum wind speed	√	√			
Short circuit	50 % of the maximum wind speed, and not less than 15 m/s	√	√		√	
Earthquake	25 % of the maximum wind speed	√	√			Relevant seismic intensity

Annex B (informative)

Specification of UHV AC equipment and conductor

Voltage specification, short-circuit current specification, noise specification, and surge arrester specification of UHV AC equipment and conductor are respectively given in Table B.1, Table B.2, Table B.3 and Table B.4.

Table B.1 – UHV voltage specification

Voltages in kV (r.m.s. value)

U_s	U_m
1 000	1 100
1 150	1 200
Key U_s Nominal system voltage U_m Highest voltage for equipment	

Table B.2 – Specification of UHV short-circuit current

Highest voltage for equipment kV (r.m.s. value)	1 100	1 200
DC time constant ms	120	120
Peak factor	2,7	2,7

Table B.3 – Noise specification

Category	China	Japan	India
Highest voltage for equipment kV (r.m.s. value)	1 100	1 100	1 200
Transformer (continuous noise) dB(A)	75	65/70/75 ^a	75
Shunt reactor (continuous noise) dB(A)	75	---	80
Outdoors SF ₆ circuit breaker (non-continuous noise) dB(A)	110	---	110
^a The noise limitation of 1 100 kV transformer in Japan depends on the location of the substation.			

Table B.4 – Surge arrester specification applied in different countries

Category	China	Japan	India
Highest voltage for equipment kV (r.m.s. value)	1 100	1 100	1 200
Nominal system voltage kV (r.m.s. value)	1 000	1 000	1 150
Rated voltage kV (r.m.s. value)	828	826	850
Continuous operating voltage kV (r.m.s. value)	638	635	723
Nominal discharge current kA	20	20	20
Steep wave impulse residual voltage under 1/10 μ s, 20 kA kV (peak value)	$\leq 1\,782$	$\leq 1\,780$ (under 1/2,5 μ s)	-----
Lightning impulse residual voltage under 8/20 μ s, 20 kA kV (peak value)	$\leq 1\,620$	$\leq 1\,620$	1 700
Switching impulse residual voltage under 30/60 μ s, 2 kA kV (peak value)	$\leq 1\,460$	$\leq 1\,400$ (under 30/80 μ s, 1 kA)	1 500
Reference voltage under direct-current 8 mA kV (peak value)	$\geq 1\,114$	$\geq 1\,050$ (under 12 mA)	-----

Annex C (informative)

1 000 kV outdoor overhead flexible conductor for UHV AC substations in China

C.1 General

The first 1 000 kV UHV AC substation – Changzhi 1 000 kV substation – is taken as an example. It is preliminarily determined to use four-bundled conductors, and corona check is made on three types of four-bundled conductors – ACSR-1, ACSR-2, ACSR-3.

ACSR-1: four-bundle thermal resistant diameter-expanded hollow aluminium alloy wire, the diameter of sub-conductor is 70 mm and the aluminium conductor's cross-sectional area of the sub-conductor is 1 600 mm².

ACSR-2: four-bundle extra light steel reinforced aluminium wire, the diameter of the sub-conductor is 51 mm and the aluminium-conductor's cross-sectional area of the sub-conductor is 1 400 mm².

ACSR-3: four-bundle diameter-expanded hollow aluminium wire, the diameter of sub-conductor is 51 mm and the aluminium-conductor's cross-sectional area of the sub-conductor is 600 mm².

The ACSR-1 is finally selected and its bundle spacing and sub-span are optimized. The process is introduced as follows:

C.2 Environmental conditions

- a) Ambient temperature
maximum temperature: +37 °C
minimum temperature: –27,2 °C
- b) Ice coating thickness: 16 mm
- c) Wind speed: Average maximum wind speed 31 m/s
- d) Altitude: < 1 000 m
- e) Site pollution severity class: d

C.3 Current-carrying capacity and thermal stability check

C.3.1 Current-carrying capacity check

Proximity effect of bundle conductor lead to a decrease of current-carrying capacity, the actual current-carrying capacity of bundle conductor can be calculated as follow

$$Z = 4\pi\lambda \left(\frac{d_0}{2} \right)^2 \frac{s}{\left(\frac{d_0}{2} \right)^2 (\rho + 1)} \quad (\text{C.1})$$

$$B = \left\{ 1 - \left[1 + \left(1 + \frac{1}{4} Z^2 \right)^{\frac{1}{4}} + \frac{10}{20 + Z^2} \right] \times \frac{Z^2 d_0}{(16 + Z^2) d} \right\}^{\frac{1}{2}} \quad (\text{C.2})$$

$$I = n I_{xu} \frac{1}{\sqrt{B}} \quad (\text{C.3})$$

where

B is the proximity effect coefficient;

S is the subconductor cross section (mm²);

d_0 is the subconductor diameter (cm);

ρ is the twist ratio of subconductor;

n is the number of subconductors;

λ is the conductivity of subconductors;

d is the spacing of subconductors (cm);

I_{xu} is the subconductor current (A).

The actual current-carrying capacities of ACSR-1, ACSR-2 and ACSR-3 are listed in Table C.1

Table C.1 – Current-carrying capacity of bundle conductor

Category	ACSR-1	ACSR-2	ACSR-3
Proximity effect coefficient	1,064	1,226	1,163
Actual current-carrying capacity A	>10 000	5 533	3 695

C.3.2 Thermal stability check

Minimum cross-section for thermal stability is

$$A = I \frac{\sqrt{t}}{C} \quad (\text{C.4})$$

where

A is the cross section (mm²);

I is the fault current (kA, r.m.s. value);

t is the current duration (s);

C is thermal stability coefficient of conductor.

The remarkable thing is that the fault current should contain the DC component. For this UHV AC substation, $I = 63$ kA, $t = 0,35$ s, $C = 87$, and the A is 428,4 mm², which is far less than the cross-section of the listed three kinds conductor.

C.4 Determination of bundle spacing

C.4.1 General

Larger bundle spacing increases the equivalent radius and average capacitance of the conductor, the average electric field strength around the conductor thus increases and the maximum electric field strength has a tendency to increase. However, the increased bundle spacing decreases the additional influence coefficient of the electric fields around the sub-conductors, and the maximum electric field strength around the conductor thus has a tendency to decrease. If the former effect is greater than the latter, the electric field strength around the conductor increases; if the former effect is less than the latter, the electric field strength around the conductor decreases. As a result, it is necessary to determine the best bundle spacing so as to minimize the electric field strength around the conductor.

C.4.2 Calculation of maximum electric field strength around conductor

The potential coefficient matrix can be used to calculate the electric field strength E_m around a conductor:

$$E_m = \frac{\sqrt{2}CU_L}{2\pi\epsilon r_0\sqrt{3}} = 0,00147 \frac{CU_L}{r_0} \quad (C.5)$$

where

E_m is the maximum electric field strength (MV/m);

U_L is the line voltage (kV);

C is the phase conductor positive sequence capacitance (pF/m);

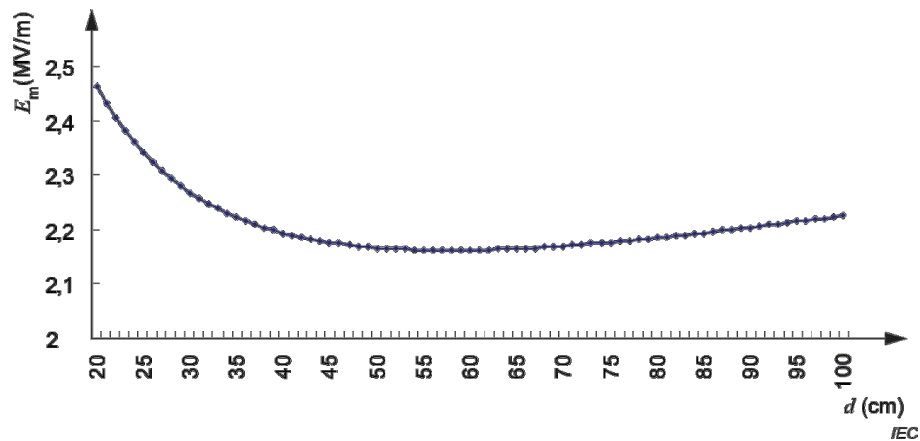
r_0 is the conductor radius (cm);

ϵ is the dielectric constant (F/m).

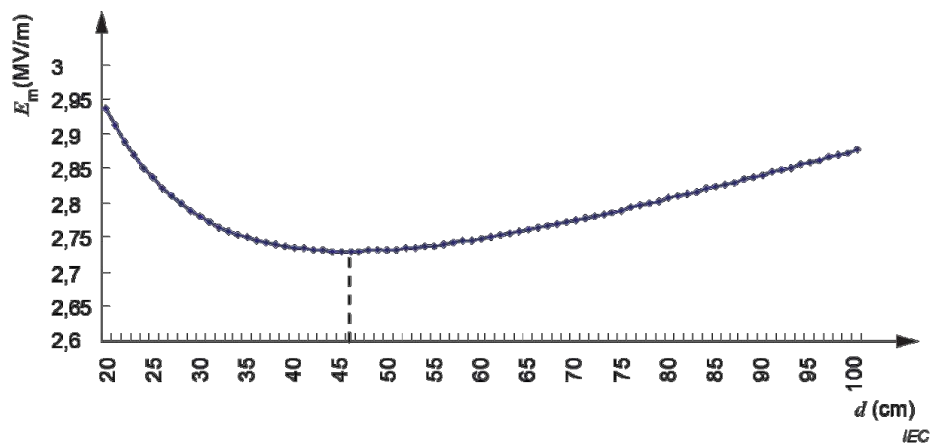
When three phase conductors are arranged horizontally in parallel, Maxwell's potential coefficient matrix can be used to accurately calculate the capacitance C of the middle conductor and thus determine the maximum electric field strength around the conductor accurately.

The relationship between the maximum electric field strength and the bundle spacing is calculated for the ACSR-1, ACSR-2 and ACSR-3 type steel core stranded aluminium conductors, and the results are shown in Figure C.1.

.



a) ACSR-1 conductor



b) ACSR-2 and ACSR-3 conductors

Figure C.1 – Relationship between maximum electric field strength and bundle spacing

According to the results, the ACSR-1 conductor with a bundle spacing of 60 cm has the minimum electric field strength.

C.5 Corona inception voltage

The corona inception voltage of a conductor should be greater than the maximum operating voltage at its corresponding installation position. The corona inception voltage of a split conductor can be calculated as follows:

$$U_0 = 84m_1m_2K\delta^{\frac{2}{3}} \frac{mr_0}{K_0} \left(1 + \frac{0,301}{\sqrt{r_0\delta}} \right) \lg \frac{a_{jj}}{r_d} \quad (\text{C.6})$$

$$\delta = \frac{2,895p}{273+t} \times 10^{-3} \quad (\text{C.7})$$

$$K_0 = 1 + \frac{r_0}{d} 2(n-1) \sin \frac{\pi}{n} \quad (\text{C.8})$$

$$t = 25 - 0.005H \quad (\text{C.9})$$

where

- U_0 is the corona inception voltage (kV);
- K is the non-uniform capacitance coefficient of the middle conductor in case of three phase conductors arranged in parallel;
- K_0 is the additional influence coefficient of the electric fields around sub-conductors;
- n is the number of sub-conductors;
- d is the spacing of sub-conductors (cm);
- m_1 is the conductor surface roughness coefficient;
- m_2 is the weather coefficient, 1,0 for sunny days and 0,85 for rainy days;
- r_0 is the conductor radius (cm);
- r_d is the equivalent radius of split conductor (cm);
- a_{jj} is the geometric mean distance between conductors;
- a is the distance between phases (cm);
- δ is the relative air density;
- p is the atmospheric pressure (Pa);
- t is the air temperature (°C);
- H is the altitude (m).

The calculation results of corona inception voltage for the ACSR-1, ACSR-2 and ACSR-3 conductors are shown in Table C.2.

Table C.2 – Corona inception voltage of conductor

Category	ACSR-1	ACSR-2	ACSR-3
Corona inception voltage, kV (r.m.s. value)	1 422	1 137	1 137

The maximum operating voltage at the cross-line position is 1 100 kV, but U/U_0 (the ratio of the maximum operating line voltage to the corona inception voltage) should be limited to less than 0,9, so as to meet the requirement of no visible corona on sunny days, that is, the corona inception voltage of the conductor should not be lower than 1 223 kV. Therefore, the ACSR-2 and ACSR-3 conductors cannot meet the requirements. It is recommended to use the ACSR-1 conductor for 1 000 kV and above cross lines.

C.6 Electric field strength on ground caused by electrostatic induction

A 1 000 kV substation has many intricate live parts, which form a complex high electric field in their surrounding space. At the same time, there are many grounded parts in the substation, such as terminal boxes, structures, etc., which have shielding effects on the electric field. The ground weeds and ground irregularities can also cause electric field distortion. All these make the electric field strength 2 m above the ground no longer evenly distributed. Moreover, if an inductive object is placed in a strong electric field, the electric field will be distorted to an extent which is related to the geometry and location of the object. As a result, it is very difficult to accurately calculate the ground electric field strength. A more accurate method is to use the finite element method for modeling calculation.

For the main transformer incoming lines of the Changzhi 1 000 kV substation, the ground electric field is affected by the conductors, insulator strings, fittings, equalizing rings and surge arrester and capacitive voltage transformer devices. A three-dimensional finite element model should be established for the whole circuit to calculate the ground electric field strength

numerically. Figure C.2 shows the plan view and sectional view of the main transformer incoming lines.

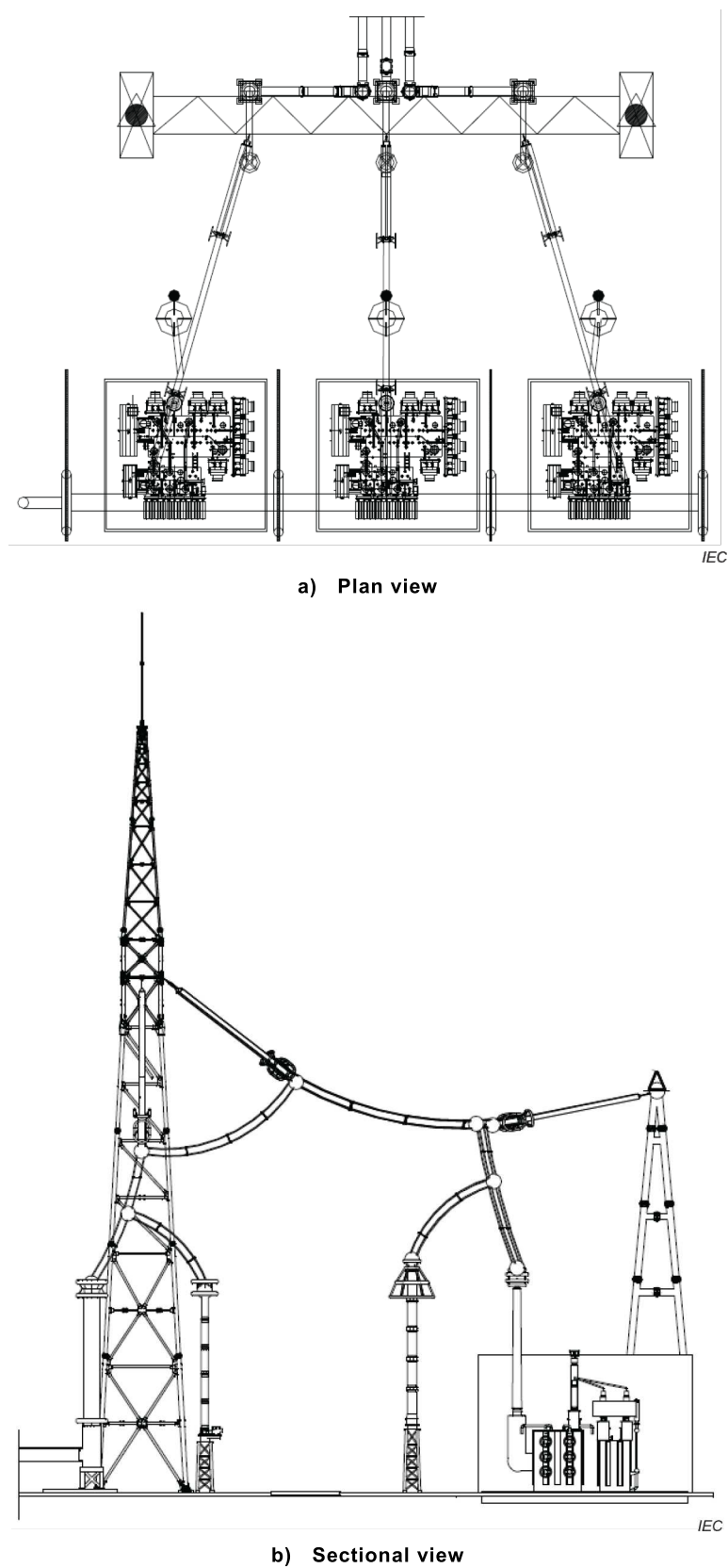


Figure C.2 – Layout plan of main transformer incoming lines

Annex D (informative)

Corona noise reduction measures of a UHV AC substation conductor under the rainy condition in Japan

D.1 Basic concept of corona noise reduction

The measure to reduce the corona noise is one of the important design issues for UHV AC substations, especially AIS and MTS. Two major causes of corona noise are radio interference (RI) which causes radio disturbance, and corona audible noise (AN).

In Japan, the regulation for RI in the AM radio band (526,5 kHz to 1 606,5 kHz) is relatively strict, and the Government Directive requires action if for radio receivers the available SN (signal-to-noise) ratio is less than 20 dB. Therefore, the design of transmission lines has been based on the SN ratio of 20 dB or above.

On the other hand, the radio noise generated in a substation is not only emitted directly to its surroundings, but also propagates through the transmission line and then is emitted from the line. Accordingly, the countermeasure for corona noise of substations focused on RI, and its suppressed level is coordinated with that of transmission lines.

The design concept of UHV AC substations will be same as that of 500 kV substations which have a lot of AIS and MTS, and have many field experiences of corona noise reduction.

Specifically, the RI electric field strength right below the 500 kV double-circuit transmission line (4-bundle conductors with ACSR-4) in rainy condition is around 60 dB [$\mu\text{V/m}$]. Accordingly, the allowable RI level of substations has been set for 60 dB [$\mu\text{V/m}$]. With regard to the AN of substations, disturbances have not been caused up to now because corona discharges are suppressed from the RI requirement, and no special considerations have been taken.

D.2 Structure design of UHV AC substation conductor

For UHV transmission lines as shown in Table D.1, 8-bundle conductors ACSR-5 have been adopted (8-bundle conductors ACSR-6 are adopted in mountainous areas).

These designs were determined taking into consideration not only RI but also AN, because they become a predominant factor of corona disturbances when the system voltage increases.

The same conductors were adopted in UHV AC substation as used in UHV AC transmission line.

The AN of 500 kV transmission lines that have long-term operating experiences is around 50 dB(A), and problems have not been caused. On the other hand, complaints have been generated in overseas transmission lines with about 55 dB(A).

Therefore the allowable level for UHV AC transmission lines was determined to be 50 dB(A) or below in light rain condition in the vicinity right under the transmission lines.

Although random noise is predominant in AN, corona hum noise (pure sound; with power frequency component and even number multiple component of power frequency) was also considered to be smaller as much as possible, because such noises often cause discomfort once they are perceived.

$G_{\max}$ shown in Figure D.1 is the maximum electric field on conductor surface, and G_p shown in Figure D.1 is that in the bottom of conductor where water droplets are formed at rain condition and from which corona is mainly generated. These values show roughly calculated ones from the following approximate equations which assume complete three-phase transposition. Therefore, the allowable level for conductors in UHV AC substation was determined to be 50 dB(A) or below in light rain condition in the vicinity right under the conductor.

Table D.1 – Estimated values of corona noise of UHV AC transmission line

Items		550 kV	1 100 kV		Reference
Circuit		Double-circuit	Double-circuit		Sub-conductor distance; 40 cm, symmetric configuration
Conductor		ACSR-4	ACSR-5	ACSR-6	
$G_{\max}$ kV/cm (r.m.s. value)		15,5	14,2	15,5	At 1,0 E (1,0 E = Maximum voltage 550 or 1 100 / $\sqrt{3}$ kV)
G_p kV/cm (r.m.s. value)		14,9	14,0	15,3	
RI dB [μ V/m]		61	60	64	Under heavy rain condition
AN dB(A)	Random- noise	48	45	50	Under light rain condition (Estimated 50 % value of the statistical distribution, rainfall; 0,5 mm/h)
	Corona- hum- noise	-	34	41	

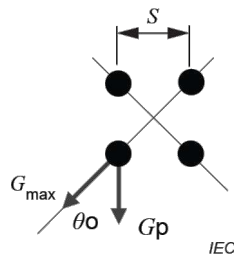


Figure D.1 – Conductor design of UHV AC substation

$$G_{\max} = \frac{V \left\{ 1 + \frac{r}{S} \cdot 2(n-1) \cdot \sin \frac{\pi}{n} \right\}}{n \cdot r \cdot \ln \left(\frac{D}{r_e} \right)} \quad (\text{D.10})$$

$$G_p = \frac{V \left\{ 1 + \frac{r}{S} \cdot 2(n-1) \cdot \sin \frac{\pi}{n} \cdot \cos \theta_0 \right\}}{n \cdot r \cdot \ln \left(\frac{D}{r_e} \right)} \quad (\text{D.2})$$

$$r_e = \sqrt[n]{n \cdot r \cdot \left\{ \frac{S}{2 \cdot \sin\left(\frac{\pi}{n}\right)} \right\}^{n-1}} \quad (\text{D.3})$$

where

- V is the applied voltage to earth;
- n is the number of sub-conductors of the bundle;
- S is the sub-conductor distance;
- D is the phase to phase distance;
- r is the radius of sub-conductor;
- r_e is the equivalent radius of the bundle.

Even under the same electric field, as the diameter and number of conductors increase, corona discharge tends to extend more easily, thus both RI and AN will also increase.

AN values are estimated ones based on actual measurement data obtained from a corona cage in the research institute of electric power industry in Japan. The values enclosed in parentheses refer to noise values when spiral wires are wound around those conductors to prevent wind noise.

D.3 Design criteria of partial discharge on UHV AC substation conductor

For busbar designs, the minimum conductor size and the number of conductors are determined to meet the RI level of 60 dB [$\mu\text{V}/\text{m}$] under rainy conditions. Regarding substation equipment and insulators to suppress the corona noise below that of the busbar by an appropriate arrangement of shielding-rings is a fundamental practice. In actual design, corona noise level varies depending on the structure of connections with conductors. Therefore, the corona test as shown in Table D.2 has been performed as a design test in UHV test stations.

Table D.2 – Design criteria of partial discharge on UHV AC substation conductors

Item	Condition	Design voltage	Design criteria
Visual corona test	Dry	1,2 E (to earth)	There should be no apparent visual corona or audible-noise
RIV (Radio-Interference Voltage) measurement test	Wet	1,0 E (to earth)	RIV: 90 dB μV or below
NOTE 1,0 E = Maximum voltage $1\,100/\sqrt{3}$ kV			

D.4 Corona noise measurement of the entrance in UHV AC test station

Since in Japan GIS technology is used for substations, live parts are limited to the entrance of transmission lines. The RI level of UHV AC transmission lines is around 60 dB [$\mu\text{V}/\text{m}$], thus this value was adopted as the allowable RI level of UHV AC substations. The same value is also applied for 550 kV. The entrance of the UHV AC test station was designed to meet this requirement.

Measurement for Corona noise level was conducted at the UHV Bushing's down-dropper in UHV test station. Table D.3 shows the result for the measurement of Corona hum noise level, under different weather condition. Noise level was measured for the entire noise frequency

spectrum as well as only for the 100 Hz component depending on power frequency (50Hz). The noise from the equipment such as transformer or GIS has been superposed on the Corona noise hum. From the result, it can be observed that the corona noise level value without the influence of equipment noise has increased due to rainy weather.

Table D.3 – Results of corona noise measurements and average value of corona noise

Corona noise in dB(A)

Weather condition	Full frequency range	100 Hz	Remarks
Light rain	54,3	46,7	Average value of data obtained at 3 measuring points
Fair weather	40,6	33,0 (37,7)	Average value of data obtained at 22 measuring points
Light rain – Fair (difference)	13,7	13,7 (9)	Values enclosed in parentheses are average values of data obtained at 3 measuring points

Annex E (informative)

Typical examples of items to be considered to select switchgear type

The term AIS (Air Insulated Switchgear) is used for switchgear whose bays consist entirely of AIS technology components. Substations containing dead-tank circuit-breakers are also considered to be AIS substations.

The term GIS (Gas Insulated Switchgear) is used for switchgear whose bays consist entirely of GIS technology components. Only external HV connections to overhead or cable lines or to transformers, reactors and capacitors can have an external insulation.

The term MTS (Mixed Technology Switchgear) is used for switchgear whose bays consist of both, AIS and GIS technology components. Switchgear, that consists of bays where some of the bays are made of AIS technology components and some of the bays are made either of GIS technology components only or of a mix of AIS and GIS technology components.

The principal technology designs for substations (their components and bays) are given in Table E.1.

Table E.1 – The principal technology designs for substations (CIGRE TB 570)

Technology design	Insulation	Insulating medium	Enclosure
AIS technology	external insulation (internal insulation can be SF ₆ , air, oil, solids or all other kinds of insulating media)	Air	No enclosure or enclosure (porcelain or composite insulators) under high voltage
GIS technology	internal and external insulation	SF ₆ or SF ₆ mixtures	Metal enclosure effectively earthed
MTS	external insulation (internal insulation can be SF ₆ , air, oil, solids or all other kinds of insulating media)	SF ₆ or SF ₆ mixtures and Air	Combination of all

Various manufacturers have developed switchgear components derived from AIS or GIS, which can be assembled in a variety of configurations to perform switchgear and control gear functions as a Mixed Technology Switchgear (MTS).

These different technologies differ in their approaches: One solution is driven by the need to optimize the investment cost, another by limited space requirements, a third by the need for redundancy and higher reliability of the single line layout.

Since the introduction of GIS into the market in 1966, the substation layout for GIS design followed almost identically the AIS design layout. With the development of compact outdoor solutions, a lot of different substation layouts were built to take advantage of compact switchgear or of switchgear with combined functions.

The drivers for the development of MTS are strictly the demands of the changing world of energy supply. The economic pressures on utilities under the conditions of deregulation demand equipment with minimum life cycle cost, high availability (which will be achieved by highly reliable products with high flexibility for installation and replacement), and compact solutions to extend as effectively as possible the already existing substations.

Because of the required dimensions and weight of UHV AC equipment, gas-insulated technologies are preferred to technical solutions. The selection of GIS and MTS depends on the actual requirements. A systematic case-specific evaluation is required for each case.

In order to select the switchgear type of AIS, MTS or GIS adopted in UHV AC substation, Table E.2 shows typical examples of items to be considered and examples to be investigated for each item.

Table E.2 – Typical examples of items to be considered to select switchgear type

Category	Examples of investigation	Remarks
Reliability	– Pollution flashover – Anti-earthquake capability – Anti-weather condition endurance	
Maintenance and operation	– Working experience – Working location (e.g., height) – Safety during operation and maintenance – Easiness of daily inspection – Easiness of replacement of equipment and components – Frequency of maintenance	
Installation	– Installation period – On-site assembly – Structure of equipments, frames and supports – SF₆ gas handling	
Layout	– Easiness of outgoing – Needs of compactness of equipment and space of land – Connection between equipments – Land acquisition – Future expansion	Manufacturer may be restricted in case of GIS for future expansion.
Circumstance	– Noise – Electromagnetic interference – Global warming impact (e.g. SF₆ gas and CO₂ usage) – Material usage quantity	Recycle or reuse may be also considered.
Construction cost	– Cost of equipments – Other costs in addition to equipments (e.g., on-site assembly, connection between equipments, testing, etc.) – Back-up (stand-by) equipment – Future expansion	
Testing / Commissioning	– Care for on-site test – Period of on-site test – Experience of commissioning	For MTS and GIS, manufacturer cares the tests for most of equipments and connections between them in factory before delivery.

Annex F (informative)

Standards related to seismic design

F.1 Typical seismic guide and standards

Table F.1 – Typical seismic guide and standards

Standard (Guide) No.	Titles
IEC 62271-207	High-voltage switchgear and controlgear-Part 207: Seismic qualification for gas-insulated switchgear assemblies for rated voltage above 52 kV
IEC TR 62271-300	High-voltage switchgear and controlgear-Part 300: Seismic qualification of alternating current circuit-breakers
IEC TS 61463	Bushings-Seismic qualification
IEC 60068-2-57	Environmental testing Part 2: Test methods Test Ff: Vibration: Time-history method
IEC 60068-3-3	Environmental testing Part 3: Guidance Seismic test methods for equipment
IEEE 693-2005	IEEE Recommended Practice for Seismic Design of Substations
IEEE 1527-2006	IEEE Recommended Practice for the Design of Flexible Bus work Located in Seismically Active Areas

F.2 Comparison of main items among the seismic standards

Table F.2 – Comparison of main items among seismic standards

Category		IEC 62271-207	IEEE 693-2005
Classification of equipment		GIS	Overall
Frequency range		0,5 Hz to 33 Hz	0,3 Hz to 33 Hz
Acceleration ratio (Vertical/Horizontal)		0,5	0,8
Damping ratio (If unclear)		2 %	2 %
Test	Input acceleration (g)	0,5, 0,25	0,5, 0,25, 0,1
	Waveform	Artificial earthquake wave or 5 cycles/beat	Artificial earthquake wave, 10 cycles/beat
	Exciting direction	Two axial simultaneously	Basically three axial simultaneously
Analysis	Static	Specified	Specified
	Static coefficient	Specified	Specified
	Dynamic	Time-history, response spectrum	Response spectrum
	Exciting direction	Two axial	Basically three axial

Bibliography

IEC 60068-2-57, *Environmental testing – Part 2-57: Tests – Test Ff: Vibration – Time-history and sine-beat method*

IEC 60099-4, *Surge arresters – Part 4: Metal-oxide surge arresters without gaps for a.c. systems*

IEC 60099-5, *Surge arresters – Part 5: Selection and application recommendations*

IEC 62271-112:2013, *High-voltage switchgear and controlgear – Part 112: Alternating current high-speed earthing switches for secondary arc extinction on transmission lines*

IEC 62772, *Composite hollow core station post insulators for substations with a.c. voltage greater than 1 000 V and d.c. voltage greater than 1 500 V – Definitions, test methods and acceptance criteria*

IEEE 1862-2014, *IEEE recommended practice for overvoltage and insulation coordination of transmission systems at 1000 kV AC and above*

IEEE C37.1-2007, *IEEE Standard for SACDA and Automation Systems*

IEEE Std 80-2000, *Guide for Safety in AC Substation Grounding*

IEEE Std 605-2008, *IEEE Guide for Bus Design in Air Insulated Substations*

IEEE 693-2005, *IEEE recommended practice for seismic design of substations*

IEEE 1527-2006, *IEEE Recommended Practice for the Design of Flexible Bus work Located in Seismically Active Areas*

CIGRE TB 390, *Evaluation of different switchgear technologies (AIS, MTS, GIS) for rated voltages of 52 kV and above; Working Bodies: WG B3.20 – SC: B3 – Year: 2009*

CIGRE TB 400, *Technical requirements for substations exceeding 800 kV; Working Bodies: WG B3.22 – SC: B3 – Year: 2009*

CIGRE TB 456, *Background of Technical Specifications for Substation Equipment exceeding 800 kV AC; Working Bodies: WG A3.22 – SC: A3 – Year: 2011*

CIGRE TB 519, *Very Fast Transient Overvoltages (VFTO) in Gas-Insulated UHV Substations; Working Bodies: WG D1.03 – SC: D1.03 – Year: 2012*

CIGRE TB 542, *Insulation Coordination for UHV AC Systems; Working Bodies: WG C4.306 – SC: C4 – Year: 2013*

CIGRE TB 562, *Field tests for UHV substations; Working Bodies: WG B3.29 – SC: B3 – Year: 2013*

CIGRE TB 570, *Switching Phenomena for EHV and UHV Equipment; Working Bodies: WG A3.28 – SC: A3 – Year: 2014*

CIGRE TB 585, *Circuit Configuration Optimization; Working Bodies: JWG B3/C1/C2.14- SC: B3, SC C1, SC C2 – Year: 2014*

CIGRE Brochure 362, *Technical requirements for substation equipment exceeding 800 kV AC*, pp.185 – 190

ICNIRP Guidelines for limiting exposure to time-varying electric and magnetic fields (1 Hz-100 kHz)

ASCE 7-16, *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch